Taxing Automation? Redistribution across Skills and Generations *

Shinnosuke Kikuchi† Sagiri Kitao‡

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Abstract

What are the welfare effects of automation-induced labor market polarization? Can an automation tax mitigate its distributional consequences? We develop an overlapping generations model with occupational mobility, occupation-specific human capital, and task-based production, in which equipment displaces routine labor. Calibrating automation to its estimated wage effects since the early 1980s, we find that automation raises average welfare across generations but generates substantial heterogeneity. Low-skilled workers experience small welfare losses, while high-skilled workers gain up to 10% in consumption equivalence. An automation tax redistributes welfare across skills and generations, but the tax rate that maximizes social welfare is 8%, below the status quo of 13%, and the tax cut widens wage inequality, benefiting the high-skilled at the expense of the low-skilled. The low-skilled and the generations alive in 1980 prefer rates near 40%, whereas the high-skilled and future generations prefer zero.

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†UC San Diego. Email: shinnkikuchi@gmail.com

‡Asian Development Bank Institute (ADBI) and National Graduate Institute for Policy Studies (GRIPS). Email: sagiri.kitao@gmail.com

1 Introduction

Over the past several decades, employment and wage growth in middle-skilled occupations have lagged behind those in low- and high-skilled occupations in the U.S. Industrial robots and technological innovations replaced many routine tasks that had been performed by middle-skilled workers. Acemoglu and Restrepo (2022) show that automation accounts for most of the increases in wage inequality across demographic groups in the U.S. since the 1980s.

This paper studies the welfare effects of automation-induced labor market polarization across skill groups and generations and then examines whether an automation tax could have mitigated its distributional consequences. Individuals respond to automation by adjusting occupations, labor force participation, consumption, and saving. We quantify the welfare effects of automation, and of a tax on net equipment income at various rates, for generations alive in 1980 and subsequent generations, distinguishing individuals by skill level.

We build an overlapping generations model in which individuals make these decisions over the life cycle. Each individual faces idiosyncratic productivity shocks, survival risk, and a borrowing constraint. Individuals accumulate occupation-specific human capital and lose it when they switch occupations. Switching may be worthwhile for a young worker but not for an older worker, who has more human capital to lose and fewer years to re-accumulate it in a new occupation. A switch that involves a temporary wage decline is more costly for an individual with limited buffer-stock savings who is unable to smooth consumption. Production uses abstract, routine, and manual labor, equipment, and other capital. In a task-based framework, equipment displaces routine labor: each task is assigned to the factor that can perform it at a lower cost. Wages and the share of tasks performed by equipment adjust in response to automation and to the tax on net equipment income.

We use the Current Population Survey (CPS) to estimate occupation-specific wages since the early 1980s and life-cycle profiles of human capital by education and occupation, and the Panel Study of Income Dynamics (PSID) to estimate idiosyncratic productivity shocks. We calibrate the initial economy to occupational distributions, occupational mobility, and non-employment in the early 1980s, as well as capital income shares and depreciation rates in the U.S. private business sector. We calibrate automation to match its effects on the wages of low- and high-skilled workers from 1980 to 2016 predicted by Acemoglu and Restrepo (2022). In the baseline economy, paths of occupational productivity and equipment efficiency, together with automation, account for the occupation-specific wages estimated from the CPS.

We quantify the welfare effects of automation relative to an economy in which technology remains at its 1980 level, measured in terms of consumption-equivalent variation. Automation hurts the low-skilled and benefits the high-skilled. Among individuals aged 25 in 1980, the consumption-equivalent variation is -0.71% for the low-skilled and 3.72% for the high-skilled. The welfare gain for the high-skilled is larger for later generations and reaches 10.45% for those entering in 2013. The welfare loss for the low-skilled also increases for later generations, to -0.91% for entrants in 2100, while individuals near retirement in 1980 are hardly affected. Although automation also raises non-employment, it improves welfare on average in all generations, as the gains of the high-skilled dominate the losses of the low-skilled.

We then ask whether an automation tax can mitigate these distributional effects. We change the tax on net equipment income from the status quo of 13% to an alternative constant rate starting in 1981, with uniform lump-sum transfers to offset the change in tax revenue. A higher tax raises the routine wage relative to the abstract wage, benefiting the low-skilled. The optimal tax rate that maximizes welfare across current and future generations is 8%, lower than the status quo rate of 13%. This reduction widens the wage gap between low- and high-skilled individuals.

Preferences over the tax rate differ sharply across skill groups and generations. The tax rates that maximize welfare for the low-skilled and the generations alive in 1980 are 38% and 43%, respectively, whereas the high-skilled and future generations prefer a zero tax rate. The generations alive in 1980 benefit from a higher tax mainly through the lump-sum transfers received by older individuals. When transfers are restricted to workers, the generations alive in 1980 lose at tax rates of 20%, 30%, and 40%.

Literature. First, this paper contributes to the literature on automation and labor market polarization (Autor et al. 2003; Acemoglu and Autor 2011; Autor and Dorn 2013; Acemoglu and Restrepo 2018; Bárány and Siegel 2018; Acemoglu and Restrepo 2022). These studies develop the task framework and document how computerization, automation, and structural change have shifted employment and wages across occupations and skills. We translate the wage effects estimated by Acemoglu and Restrepo (2022) into welfare effects: automation improves welfare on average, but the welfare loss of the low-skilled is small relative to their wage loss, whereas the welfare gain of the high-skilled is close to their wage gain.[1]

[1]For entrants in 2018, who face the full wage effects over their working lives, the consumption-equivalent variation is -0.83% for the low-skilled, whose wages change by -3.94 log points, and 10.29% for the high-skilled, whose wages change by 12.12 log points.

Second, this paper contributes to quantitative studies of how automation affects workers over the life cycle and across generations. Routine workers respond to automation by switching occupations or leaving the labor force (Cortes 2016; Cortes et al. 2017), and quantitative models study this reallocation (Dvorkin and Monge-Naranjo 2019; Jaimovich et al. 2021). In overlapping generations models, technological change affects generations differently: older incumbents adjust less (Adão et al. 2024), whereas losses can fall on future generations, because older generations own the capital and skills that machines complement (Sachs and Kotlikoff 2012) or because lower wages reduce saving (Benzell et al. 2018).[2] We compute welfare effects for individuals of all ages in 1980 and every future entry cohort, revealing different scopes for adjustment by age. Older workers respond less to automation, and those near retirement in 1980 are hardly affected because they have few working years left to be exposed to it. In contrast, the welfare losses of the low-skilled and the gains of the high-skilled are largest for younger and future generations.

Finally, this paper contributes to the literature on taxing automation (Acemoglu et al. 2020; Guerreiro et al. 2022; Beraja and Zorzi 2025). These studies provide arguments for taxing automation based on workers' slow adjustment to technological change or the lower taxation of capital relative to labor in the U.S. We evaluate the tax in a quantitative general equilibrium model in which individuals face borrowing constraints and costly occupational mobility over the life cycle. We find that the tax has significant redistributive effects across skill groups and generations, although the aggregate effects are minimal. Changing the rate raises social welfare by at most 0.03%, while a higher tax benefits the low-skilled through higher routine wages and older individuals alive in 1980 through lump-sum transfers, at the expense of the high-skilled and future generations. The paper also relates to studies of optimal taxation in life-cycle models with incomplete markets (Conesa et al. 2009; Heathcote et al. 2020).

Section 2 presents the data and labor market trends. Section 3 presents the model, and Section 4 discusses calibration and the initial economy. Section 5 presents the welfare effects of polarization, from automation only and from full polarization, and Section 6 presents the automation tax.

[2] Cociuba and MacGee (2018) find that the adverse effects of sectoral reallocation are concentrated on young workers. Heathcote et al. (2010) compute welfare for cohorts entering the labor market as wage inequality rises. Moll et al. (2022) show that automation also raises income and wealth inequality through higher returns to wealth.

2 Data and Trends in the Labor Market

We use the Current Population Survey (CPS) to study occupational distributions and wage profiles, and the Panel Study of Income Dynamics (PSID) to estimate idiosyncratic productivity shocks.

2.1 Data

Selection of Individuals. We use employment data for male and female household heads from 1983 to 2019. Figure 1 uses data for individuals aged 30–59. Figure 2 shows occupation shares at each age from 25 to 79, pooling observations from 1983 to 1985. Low-skilled individuals have at most a high school education, and high-skilled individuals have at least some college education. We start the analysis of occupations in 1983 because of the change in occupational categories before and after 1983, as documented in Kambourov and Manovskii (2013).

Employment Status and Occupational Distributions. We use the 1990 occupation codes of the Census and categorize occupations as abstract, routine, and manual, following Acemoglu and Autor (2011). Abstract occupations include managers, professional specialty workers, and technicians. Routine occupations include sales, administrative support, construction, production, and transportation. Manual occupations include personal services, cleaning, food preparation, and protective services. We exclude farming occupations and individuals whose occupation or employment status cannot be classified.

The frequency of our model is annual. To account for unemployment of a short duration, we classify individuals unemployed for no more than 13 weeks as employed when their previous occupation is observed. Other unemployed individuals and those not in the labor force are in non-employment.

Wage Data. For the estimation of wages, we use CPS household heads aged 25–64 from 1983 to 2018 who work 50–52 weeks and report five or more weekly hours. Years are CPS survey years, and earnings refer to the previous calendar year. Hourly wage is the sum of wage and business income, divided by 52 times weekly hours. Income is deflated by the CPI-U of the income year, in 1999 dollars. Following Baum-Snow and Neal (2009), we restrict hourly wages to the range from 1.50 to 99.80 in 1999 dollars.

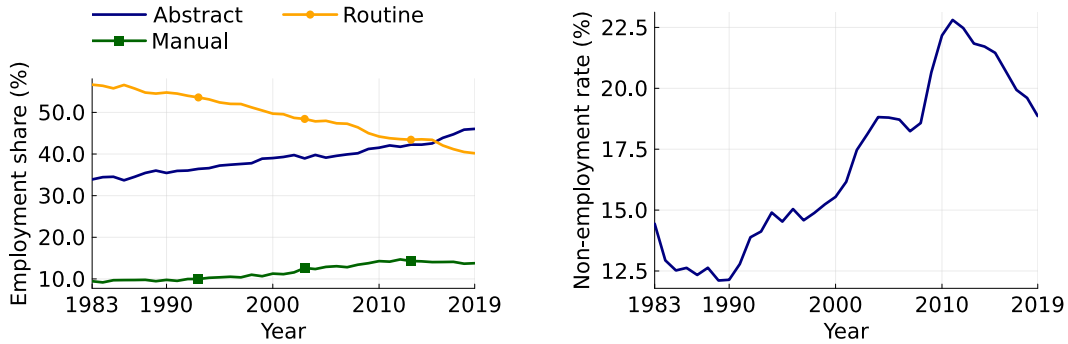
The PSID sample includes male and female household heads aged 30–59 from 1983 to 1996, employed in abstract, routine, or manual occupations. High-skilled individuals have 13 or more years of completed education. We use the core sample and the same hourly wage range.

2.2 Polarization in the Labor Market

Figure 1 shows the occupation shares over time. The share of routine occupations among employed individuals falls from 56.7% in 1983 to 40.2% in 2019. During the same period, the share of abstract occupations rises from 33.9% to 46.0%, and the share of manual occupations rises from 9.4% to 13.8%. Non-employment rises from 14.4% to 18.9%.

Figure 1: Employment Share and Non-employment Rate

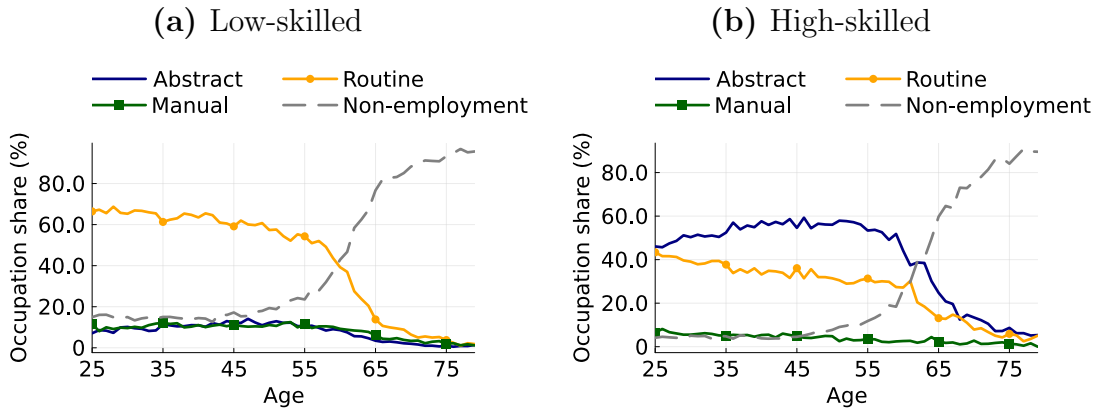
(a) Employment Share by Occupation **(b)** Non-employment Rate



Notes: This figure shows employment shares and the non-employment rate from 1983 to 2019. The data are from the Current Population Survey (CPS). The sample includes all household heads aged 30–59. Panel (a) shows shares among individuals employed in abstract, routine, or manual occupations. Panel (b) shows non-employment as a share of all individuals in the sample. Both panels use person weights. Individuals unemployed for no more than 13 weeks are in employment when their previous occupation is observed.

Figure 2 shows occupation shares over the life cycle by skill group. Routine occupations account for the largest share among low-skilled individuals throughout most of their working lives, while abstract occupations account for the largest share among high-skilled individuals. Non-employment rises at older ages. At age 65, the non-employment rate is 76.7% for the low-skilled and 59.7% for the high-skilled.

Figure 2: Occupation by Age and Skill



Notes: This figure shows occupation shares by age and skill group in 1983–1985. The sample includes all household heads in the CPS, aged 25–79. Low-skilled individuals have at most a high school education; high-skilled individuals have at least some college education. Shares include non-employment and use person weights. Occupations are abstract, routine, or manual; individuals unemployed for no more than 13 weeks are in employment when their previous occupation is observed.

3 Model

In this model, individuals choose consumption, saving, labor force participation, and occupations. Equipment can replace routine labor in production tasks. Tax rates and technology are exogenous in the baseline economy. The model frequency is annual.

3.1 Individuals

Demographics. Individuals enter the economy at age 25 ($j = 1$) with zero initial assets, $a_1 = 0$, and can live up to age 85 ($J = 61$). Let ϕ_{j+1} denote the conditional probability that an individual aged j remains alive until age $j + 1$, with $\phi_{J+1} = 0$. There is no bequest motive. Accidental bequests are confiscated by the government.

Individuals are born with a skill type $s \in \{L, H\}$, which represents education prior to entry into the labor market. Skill does not change over the life cycle. Let $N_{j,s,t}$ denote the population of age j and skill s at time t. Given the number and skills of entrants, the population evolves according to

$$N_{j+1,s,t+1} = \phi_{j+1} N_{j,s,t}. \tag{1}$$

Endowment, Occupation and Human Capital. Let $o \in \{A, R, M, N\}$ denote an individual's previous occupation: abstract, routine, manual, or non-employment. Entrants choose among the three working occupations. After entry,

workers can remain in their occupation, move to another occupation, or leave the labor force. Non-employment is an absorbing state.

Let o' denote the current occupation choice. At the start of each period, an individual draws a vector of idiosyncratic productivity shocks $\boldsymbol{\eta} = (\eta_A, \eta_R, \eta_M)$. These shocks are independent across occupations and over time. We use $\log \eta_{o'} \sim \mathcal{N}(0, \sigma_\eta^2)$ for each occupation, with the same distribution for both skills. Earnings in occupation o' are

$$y_t(o') = w_{o',t} h_{o'} \eta_{o'} \quad (o' \in \{A, R, M\}), \qquad y_t(N) = 0, \tag{2}$$

where $w_{o',t}$ is the occupation-specific wage and $h_{o'}$ is human capital after the current occupation decision.

Workers accumulate occupation-specific human capital on the job. Entrants begin with human capital $\widehat{h}_{1,s,o'}$ in occupation o'. Let h be the human capital from the previous period, $g^h_{j,s,o}$ its growth rate from age j to $j+1$, and $\widehat{h}_{j,s,o'} = \widehat{h}_{1,s,o'} \prod_{k=1}^{j-1}(1 + g^h_{k,s,o'})$ the human capital at age j of a worker who has remained in occupation o' since entry. For workers aged $j \geq 2$,

$$h_{o'} = \begin{cases} (1 + g^h_{j-1,s,o})h, & o' = o \in \{A, R, M\}, \\ \min\{\widehat{h}_{1,s,o'}, \widehat{h}_{j,s,o'}\}, & o' \neq o,\ o' \in \{A, R, M\}, \\ 0, & o' = N. \end{cases} \tag{3}$$

Individuals who change occupations lose their accumulated occupation-specific human capital and start from the entry level in the new occupation, or the age-specific level if it is lower.

Preferences. Individuals derive utility from consumption and disutility from labor participation. Period flow utility from consumption is $u(c) = c^{1-\gamma}/(1-\gamma)$, where γ is the curvature parameter. β denotes the subjective discount factor. Individuals incur a utility cost from work of $B_s + \kappa_{s,o'}$. $\kappa_{s,o'}$ represents the occupation-specific component and we normalize $\kappa_{s,R} = 0$. Individuals moving from occupation o to o' incur an additional utility cost $c^s_{oo'}$, with $c^s_{oo} = 0$. Costs of leaving the labor force are zero.

Social Security and Transfers. Individuals receive a constant social security benefit ss once they reach the eligibility age j_{ss}. The government taxes labor earnings at rate $\tau_{I,t}$. Individuals receive a uniform lump-sum transfer T_t, which can be negative. Let r^H_t denote the household return after capital income taxes. It equals the fixed after-tax return r except at the date of an unanticipated change

(Section 3.3). Resources before means-tested transfers are

$$b_t(o') = (1 + r_t^H)a + (1 - \tau_{I,t})y_t(o') + ss\,\mathbf{1}\{j \geq j_{ss}\} + T_t, \tag{4}$$

where a denotes assets at the start of the period. The means-tested transfer $CF_t(o') = \max\{0, \underline{c} - b_t(o')\}$ raises resources to the consumption floor $\underline{c}$. Social security and the consumption floor are constant over time.

Life-cycle Problem. A state vector at the start of a period is $x = (j, a, s, o, h)$. An individual draws the productivity shocks $\boldsymbol{\eta}$, chooses an occupation o' in the set of feasible choices $\mathcal{D}(o)$, and then chooses consumption and saving. The set of feasible choices is $\mathcal{D}(o) = \{A, R, M, N\}$ for a working origin and $\mathcal{D}(N) = \{N\}$. At entry it is $\{A, R, M\}$, and there is no mobility cost. Given the productivity shocks and the occupation o', the value from consumption and saving is

$$\begin{aligned} W_t(x, o', \boldsymbol{\eta}) = \max_{c, a' \geq 0} \{u(c) + \beta\phi_{j+1} V_{t+1}(j+1, a', s, o', h_{o'})\}, \\ c + a' = b_t(o') + CF_t(o'), \qquad c > 0, \end{aligned} \tag{5}$$

where V_t is the expected value at the start of a period, defined below. At the last age, the continuation value is zero.

Occupational Choice. The value of choosing o', net of participation and mobility costs, is

$$z_t(x, o', \boldsymbol{\eta}) = W_t(x, o', \boldsymbol{\eta}) - (B_s + \kappa_{s,o'})\mathbf{1}\{o' \neq N\} - c^s_{oo'}. \tag{6}$$

The individual chooses o' to maximize $z_t(x, o', \boldsymbol{\eta}) + \nu_{o'}$, where the preference shocks $\nu_{o'}$ are independent across choices and over time and follow a centered type-I extreme-value distribution with scale $1/\theta_s$. The choice probabilities and the expected value then take the logit form of McFadden (1973):

$$\Pr_t(o' \mid x, \boldsymbol{\eta}) = \frac{\exp\{\theta_s z_t(x, o', \boldsymbol{\eta})\}}{\sum_{k \in \mathcal{D}(o)} \exp\{\theta_s z_t(x, k, \boldsymbol{\eta})\}}, \tag{7}$$

$$\begin{aligned} V_t(x) &= \mathbb{E}\left[\max_{o' \in \mathcal{D}(o)} \{z_t(x, o', \boldsymbol{\eta}) + \nu_{o'}\}\right] \\ &= \mathbb{E}_{\boldsymbol{\eta}}\left\{\frac{1}{\theta_s} \log\left[\sum_{o' \in \mathcal{D}(o)} \exp\{\theta_s z_t(x, o', \boldsymbol{\eta})\}\right]\right\}. \end{aligned} \tag{8}$$

A higher θ_s means a larger response of occupation shares to differences in values; we use the same θ_s at entry and at later ages. Switching costs with extreme-value

preference shocks are standard in models of labor mobility across sectors and occupations (Artuç et al. 2010; Dix-Carneiro 2014; Dvorkin and Monge-Naranjo 2019).

3.2 Production and Automation

The final good is the numeraire. Competitive firms combine other capital $K_{O,t}$, manual labor $L_{M,t}$, and a composite X_t of abstract labor and routine tasks:

$$Y_t = \left(\frac{K_{O,t}}{\alpha}\right)^{\alpha} \left(\frac{L_{M,t}}{\mu_t}\right)^{\mu_t} \left(\frac{X_t}{1-\alpha-\mu_t}\right)^{1-\alpha-\mu_t}, \tag{9}$$

$$X_t = \left[A_{A,t} L_{A,t}^{\frac{\sigma-1}{\sigma}} + A_{R,t} Z_t^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}. \tag{10}$$

$L_{o,t}$ is effective labor in occupation o, $A_{A,t}$ and $A_{R,t}$ are production coefficients, and σ is the elasticity of substitution between abstract labor and routine tasks. Other capital receives the share α of output and manual labor the share μ_t. Manual labor enters with a Cobb-Douglas weight rather than in a CES aggregate with abstract labor and routine tasks, so automation does not have the same effect on the demand for manual and abstract labor: abstract labor is combined with routine tasks within X_t at elasticity σ, whereas manual labor gains from automation only through higher output, at the constant share μ_t.[3] Routine production uses a continuum of tasks $i \in [0,1]$:

$$Z_t = \left[\int_0^1 q_t(i)^{\frac{\zeta-1}{\zeta}}\, di\right]^{\frac{\zeta}{\zeta-1}}, \tag{11}$$

where ζ is the elasticity of substitution across tasks.

Equipment and routine labor can be used in every task. Following the routine-task specification of Acemoglu et al. (2020), labor productivity in task i is $A_{\text{task}} i^{\chi}$. Task output is

$$q_t(i) = A_{E,t}\psi_t(i)k_t(i) + A_{\text{task}} i^{\chi} \ell_t(i), \tag{12}$$

with $\int_0^1 k_t(i)\, di = K_{E,t}$ and $\int_0^1 \ell_t(i)\, di = L_{R,t}$. Here $A_{E,t}$ is the efficiency of equipment and $\psi_t(i)$ is the productivity of equipment across tasks:

$$\psi_t(i) = \begin{cases} m_t, & 0 \le i \le \iota_0, \\ e^{\Upsilon_t}(i/\iota_0)^b, & \iota_0 < i \le 1, \end{cases} \qquad m_t \ge 1,\ \Upsilon_t \ge 0,\ 0 < b < \chi, \tag{13}$$

[3] This follows Autor et al. (2003), who find that computerization complements abstract tasks and has little direct effect on manual tasks, and Autor and Dorn (2013), in which the demand for manual services rises with income. Section 6.4 reports results with manual labor in the CES aggregate: the optimal tax is unchanged, and the welfare gains by skill at that rate are close to those of the main specification.

where ι_0 is the share of tasks automated in 1980. Automation raises m_t, the productivity of equipment in tasks already automated, and Υ_t, its productivity in the remaining tasks, where labor keeps a comparative advantage because $b < \chi$. In 1980, $A_E = m = 1$ and $\Upsilon = 0$, and equipment performs only the tasks in $[0, \iota_0]$, as with $\psi \equiv 1$.

Firms assign task i to equipment if its unit cost there is below that of routine labor,

$$\frac{u_{E,t}}{A_{E,t}\psi_t(i)} \leq \frac{w_{R,t}}{A_{\text{task}} i^{\chi}}, \tag{14}$$

where $u_{E,t}$ is the rental cost of equipment. The share of automated tasks ι_t is the measure of this set, which need not be an interval. In 1980, the automated tasks are $[0, \iota_0]$, with

$$\iota_0 = \left(\frac{w_{R,1980}}{A_{\text{task}} u_{E,1980}} \right)^{1/\chi}. \tag{15}$$

The parameters A_{task}, χ, ι_0, and b are constant.

Equipment and other capital have a price of one in units of the final good; changes in the efficiency of equipment are captured by $A_{E,t}$. The government taxes net capital income at rates $\tau_{E,t}$ and τ_O. At dates when capital can adjust, rental costs satisfy

$$u_{E,t} = \frac{r}{1 - \tau_{E,t}} + \delta_E, \tag{16}$$

$$u_{O,t} = \frac{r}{1 - \tau_O} + \delta_O. \tag{17}$$

Here δ_E and δ_O are depreciation rates. Firms use labor at its marginal product, choose $K_{O,t} = \alpha Y_t / u_{O,t}$, and choose $K_{E,t}$ so that the marginal product of equipment, after allocating tasks at minimum cost, equals $u_{E,t}$. Equivalently, firms demand effective equipment $A_{E,t}K_{E,t}$ at the cost $u_{E,t}/A_{E,t}$.

3.3 Government and Equilibrium

Households hold a common portfolio of equipment, other capital, and net foreign assets. Let $\mathcal{A}_t$ be aggregate household wealth at the start of period t. Net foreign assets are

$$B_t = \mathcal{A}_t - K_{E,t} - K_{O,t}. \tag{18}$$

Define net capital income before tax as

$$\Pi_{E,t} = (Y_{K_E,t} - \delta_E) K_{E,t}, \tag{19}$$

$$\Pi_{O,t} = (Y_{K_O,t} - \delta_O) K_{O,t}, \tag{20}$$

where $Y_{K_E,t}$ and $Y_{K_O,t}$ are the marginal products of equipment and other capital. Net foreign assets earn r, so the household return satisfies

$$r_t^H \mathcal{A}_t = (1-\tau_{E,t})\Pi_{E,t} + (1-\tau_O)\Pi_{O,t} + rB_t. \tag{21}$$

At dates when capital can adjust, the marginal products equal the rental costs, so $(1-\tau_{E,t})\Pi_{E,t} = rK_{E,t}$, $(1-\tau_O)\Pi_{O,t} = rK_{O,t}$, and $r_t^H = r$. At the date of an unanticipated change in technology or policy, both capital stocks are inherited, and their realized returns, and hence r_t^H, can differ from r.

Let $Y_{L,t}$ denote aggregate labor earnings, $\mathcal{E}_t$ accidental bequests, H_t total social security and means-tested payments, and G_t government expenditures. Total population is $N_t = \sum_{j,s} N_{j,s,t}$. The government budget constraint is

$$G_t + H_t + N_t T_t = \tau_{I,t} Y_{L,t} + \tau_{E,t}\Pi_{E,t} + \tau_O \Pi_{O,t} + \mathcal{E}_t. \tag{22}$$

There is no government debt. Given tax rates, expenditures or lump-sum transfers adjust to satisfy the budget constraint.

An equilibrium includes household decisions, occupation shares, the distribution of assets and human capital, production, wages, and net foreign assets such that individuals and firms optimize, the population distribution follows household decisions and survival, and labor and goods markets clear. In particular, effective labor equals the sum of $h_{o'}\eta_{o'}$ across individuals who choose each working occupation. Goods-market clearing is given by

$$Y_t = C_t + I_{E,t} + I_{O,t} + G_t + NX_t, \tag{23}$$

$$I_{E,t} = K_{E,t+1} - (1-\delta_E)K_{E,t}, \qquad I_{O,t} = K_{O,t+1} - (1-\delta_O)K_{O,t}. \tag{24}$$

Here C_t denotes aggregate consumption. Net exports satisfy $NX_t = B_{t+1} - (1+r)B_t$. Aggregates are levels, so in a stationary state with population growth n, $NX = (n-r)B$.

4 Calibration

The initial economy represents the U.S. labor market in the early 1980s, and exogenous paths of demographics, wages, and technology move the economy from 1980. Table 2 summarizes the parameters.

4.1 Demographics

Initial Economy. Survival probabilities are constant over time. At each age, the survival probability averages the male and female probabilities in the 1990 U.S. life table, weighted by the shares of men and women among CPS household heads of that age in 1980.

The initial stationary economy uses the stable age distribution implied by the survival probabilities and the population growth rate from 1980 to 1981, with the 1980 skill shares of household heads aged 25–85 at every age.

Entrants. Entrants correspond to CPS household heads aged 25. For each skill, we fit a Hodrick–Prescott trend ($\lambda = 100$) to the log number of entrants over 1980–2020. Through 2017, entrants of each skill follow this trend, so the high-skilled share of entrants follows the CPS trend. From 2018 onward, the high-skilled share is fixed at its 2018 CPS value. The total number of entrants follows the sum of the two trends through 2020. Thereafter, the trend growth rates converge smoothly to zero by 2050 and remain zero thereafter. The final stationary economy has zero population growth and the 2018 skill shares.

4.2 Human Capital and Wages

Human Capital. We estimate

$$\log \tilde{w}_{j,s,o,t} = \pi_{j,o} + \pi_{j,s} + \omega_t + \omega_{o,t} + \rho_{t-j} + \pi_F \mathbf{1}_F + \varepsilon_{j,s,o,t} \tag{25}$$

on the CPS wage sample, where $\tilde{w}_{j,s,o,t}$ is the hourly wage of a worker of skill s in occupation o at age j in year t, $\mathbf{1}_F$ indicates women, and the age effects are constant within five-year age groups. Following Deaton (1997), the year effects, the occupation-specific year effects and the cohort effects each sum to zero and are orthogonal to a linear trend.[4] The human capital profile $\widehat{h}_{j,s,o}$ is the exponential of a cubic function of age fitted to $\hat{\pi}_{j,o} + \hat{\pi}_{j,s}$, normalized to one for low-skilled workers at age 25 in each occupation. Human capital growth is zero at age 65 and older. Figure 3(a)–(c) shows the profiles.

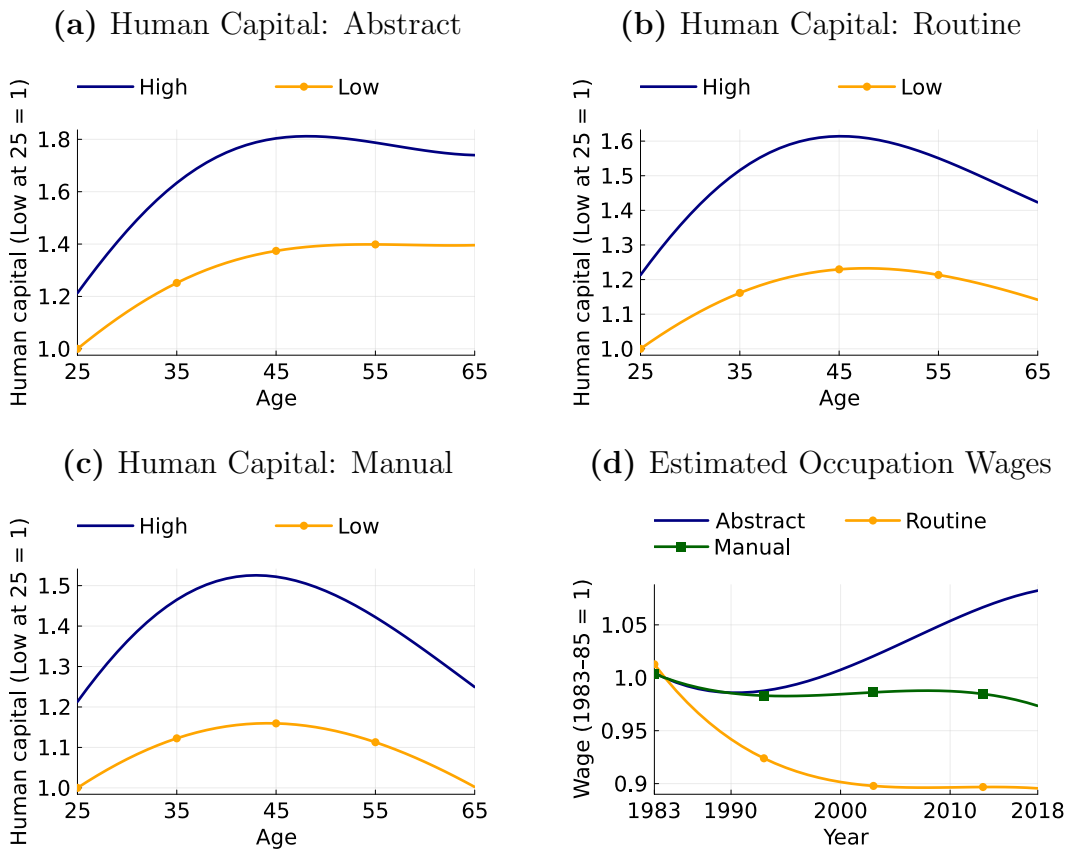
Occupation Wages. Occupation wages are weighted annual means of $\tilde{w}_{j,s,o,t}/[\widehat{h}_{j,s,o} \exp(\hat{\pi}_F \mathbf{1}_F)]$ within occupations, smoothed with occupation-specific cubic functions of time. The initial wage for each occupation is its average over 1983–1985. The common wage scale is chosen to match average earnings of one

[4] Unlike Aguiar and Hurst (2013), we also restrict the cohort effects. These restrictions identify the age profiles under an assumed separation of age effects and secular wage growth. Occupational wage changes enter the model through $w_{o,t}$.

in the initial economy. In the model, occupation wages move from their initial values to the estimated 2018 values, with linear changes in log wages over time, and remain constant thereafter. Figure 3(d) shows the estimated wages, each divided by its own initial value.

Productivity Shocks. We set $\sigma_\eta = 0.25$ by dividing the variance of annual changes in PSID log wage residuals for workers who stay in the same occupation by two and taking the square root.[5]

Figure 3: Human Capital Profiles and Occupation Wages



Notes: Panels (a)–(c) show the human capital profiles $\widehat{h}_{j,s,o}$ of high-skilled (High) and low-skilled (Low) workers by age in each occupation, normalized to one for low-skilled workers at age 25 in each occupation; human capital growth is zero at age 65 and older. Panel (d) shows estimated CPS wages from 1983 to 2018, deflated by the CPI-U, after accounting for human capital and a sex indicator. Each occupation's wage is divided by its average over 1983–1985, the value used for 1980 in the model; the model uses a path that is linear in logs between this average and the 2018 value.

[5]Residuals are from regressions, by skill, of log hourly wages on age groups and years, each interacted with occupation, and on cohort groups. Independent annual productivity draws imply a variance of $2\sigma_\eta^2$ before selection. We use this as an approximation for stayers, whose shocks are selected through occupational choice.

4.3 Preferences and Occupational Mobility

We set $\gamma = 2$, a standard value. The household return is $r = 4.21\%$, the average real yield on AAA corporate bonds over 1981–2017 used by Acemoglu et al. (2020). The subjective discount factor is calibrated jointly with the household parameters to match, at this return, the capital-output ratio of 2.7 in Conesa et al. (2009), which we target with initial household wealth relative to output. The calibrated value is $\beta = 0.97$. Household wealth differs from domestic capital by net foreign assets, which are $B_{1980}/Y_{1980} = -0.42$ in 1980. The participation costs B_s are chosen to match participation rates at age 65 for each skill type.

We use the CPS variable for occupation in the previous year to construct the transition matrices in Table 1. The sample consists of household heads aged 30–59 in 1983–1985 who work in one of the three occupations in both periods. We calibrate the six directional mobility costs $c^s_{oo'}$ for each skill type to these probabilities. The occupation-specific utility costs $\kappa_{s,A}$ and $\kappa_{s,M}$ are chosen to match occupation shares among employed individuals at age 25 in 1983–1985. Table A.1 in the Appendix reports the calibrated costs. The parameters θ_s are chosen so that the change in the routine share among entrants between the initial stationary state and a stationary state at the 2018 wages matches its change in the CPS from 1983–1985 to 2018.

Table 1: Occupation Transition Matrix

(a) Low-skilled

	Abstract	Routine	Manual
Abstract	96.75	2.63	0.61
Routine	0.55	98.89	0.56
Manual	0.48	3.47	96.05

(b) High-skilled

	Abstract	Routine	Manual
Abstract	98.20	1.60	0.20
Routine	2.36	97.19	0.46
Manual	2.71	4.29	93.00

Notes: This table shows occupation transition probabilities, in percentage terms. The sample is CPS household heads aged 30–59 in 1983–1985, employed in abstract, routine, or manual occupations in both periods. The previous occupation is on the left; the current occupation is above the probabilities. Low-skilled individuals have at most a high school education; high-skilled individuals have at least some college education. Probabilities use person weights.

4.4 Production and Technology

Initial Economy. We set $\zeta = 0.5$, the elasticity of substitution between tasks in Acemoglu and Restrepo (2022), based on Humlum (2020), and $\sigma = 0.6$, between this value and the estimate of 0.9 in Goos et al. (2014), and close to the elasticity of 0.67 between skilled labor and equipment in Krusell et al. (2000). With $\sigma < 1$, abstract labor and routine tasks are complements, and because manual labor has a Cobb-Douglas weight, automation raises the demand for abstract labor by more than the demand for manual labor. Factor shares and depreciation rates are taken from the private business sector data in the total factor productivity accounts of the U.S. Bureau of Labor Statistics (2025), using the 1987 values as proxies for our initial economy in 1980.[6] Equipment receives $s_E = 17.4\%$ of value added and other capital $\alpha = 15.8\%$, and depreciation rates are $\delta_E = 12.4\%$ and $\delta_O = 2.8\%$.

Given 1980 wages and labor, the labor share $1 - \alpha - s_E$ determines output Y, and the capital shares determine $K_E = s_E Y/u_E$ and $K_O = \alpha Y/u_O$. The manual wage bill gives $\mu_{1980} = w_M L_M/Y = 4.39\%$. Given abstract labor and routine tasks in 1980, A_A and A_R are the values at which their marginal products equal the abstract wage and the unit cost of routine tasks.

We determine the three task parameters, χ, ι_0, and A_{task}, using the long-run elasticity of substitution between equipment and routine labor, the equipment share of routine-task costs in 1980, $s_Z = u_E K_E/(u_E K_E + w_R L_R) = 28.9\%$, and the equal-cost condition at the initial automation cutoff. With the assignment of tasks fixed, equipment and routine labor substitute only across tasks, so their elasticity of substitution is ζ. When the assignment adjusts, firms also move tasks between equipment and labor, and the long-run elasticity is $\sigma_{LR} = \zeta + 1/[\chi(1-s_Z)]$; the second term is larger when the comparative advantage of labor, governed by χ, rises more slowly across tasks. We set $\chi = 1.66$ so that $\sigma_{LR} = 1.35$, the midpoint of the range implied by Karabarbounis and Neiman (2014), as in Acemoglu et al. (2020): equipment and routine labor are substitutes in the long run, although tasks are complements. Given χ, the 1980 share of automated tasks, $\iota_0 = 12.9\%$, is the cutoff consistent with s_Z: it solves $\iota_0^{-\xi} = 1 + \xi(1-s_Z)/s_Z$ with $\xi = 1 + \chi(\zeta - 1)$. Finally, A_{task} makes equipment and routine labor equally costly in the cutoff task at 1980 prices, by equation (15).

Automation. The automation technology is identified from Acemoglu and Restrepo (2022). Their predicted general-equilibrium effects of automation on log hourly wages from 1980 to 2016 (Table VIII, column 3), aggregated with 1980

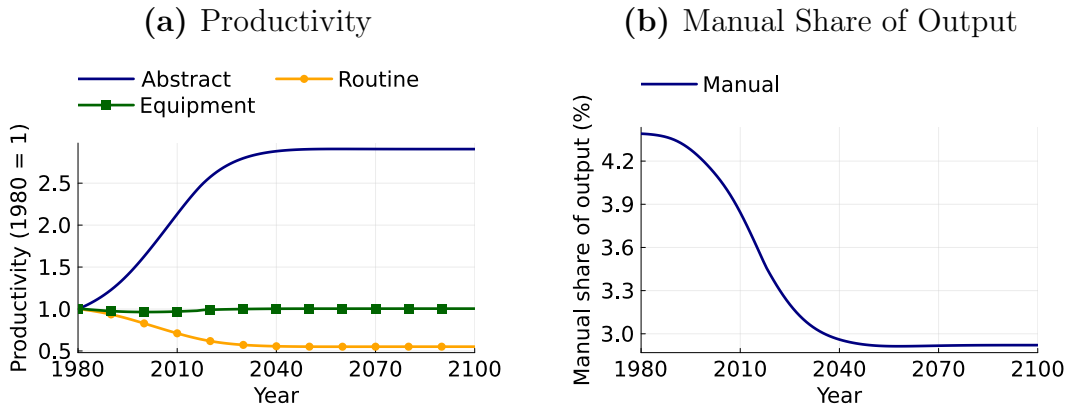
[6] Equipment comprises equipment and intellectual property products; other capital comprises structures, rental residential capital, land, and inventories. δ_E is the stock-weighted depreciation rate of equipment and intellectual property products, and δ_O that of nonresidential structures.

hours weights, are -3.94 for workers with at most a high school education and 12.12 for workers with at least some college education, measured as log changes multiplied by 100. The automation economy adds automation to technology held at its 1980 level; Section 5 compares it with constant technology. Each year, Υ_t and m_t in equation (13) are set so that the change in each skill's wage index relative to constant technology equals the predicted effect, reached linearly from 1980 to 2016 and held constant thereafter. A skill's wage index weights the log occupational wages by that skill's 1980 earnings shares across occupations. The other technology parameters remain at their 1980 values, and the slope of equipment productivity above ι_0 is $b = 0.95\chi$. To first order, as in Acemoglu and Restrepo (2022), the automation technology raises TFP by 3.7% from 1980 to 2016, close to their 3.4%. In the automation economy, the share of automated tasks rises from 12.9% in 1980 to 55.1% in 2018.

Baseline. The baseline economy takes the automation technology (Υ_t, m_t) and the estimated CPS wages as given. Each year, equipment efficiency $A_{E,t}$ sets the equipment share of output to a path that moves linearly from the 1987 BLS share, 17.4%, in 1980 to the 2018 BLS share, 19.9%, and remains constant thereafter, with $A_{E,1980} = 1$. Given the wages and household labor supply, $A_{A,t}$, $A_{R,t}$, and $\mu_t = w_{M,t}L_{M,t}/Y_t$ are the values at which firms demand the supplied labor. Figure 4 shows $A_{A,t}$, $A_{R,t}$, and $A_{E,t}$ in panel (a) and μ_t in panel (b).[7] Equipment efficiency is 0.99 in 2018, and the share of automated tasks rises from 12.9% to 21.5%. The policy experiments take the paths of $A_{A,t}$, $A_{R,t}$, μ_t, and $A_{E,t}$ and the automation technology as given.

[7] $A_{E,t}$ is a residual: equipment productivity changes mainly through the automation technology (Υ_t, m_t), and $A_{E,t}$ aligns the equipment share of output with the BLS data. Small changes suffice because, with b close to χ, equipment and routine labor have similar costs in many tasks.

Figure 4: Production Coefficients of the Baseline



Notes: This figure shows the production parameters of the baseline from 1980 to 2100. Panel (a) shows the production coefficients $A_{A,t}$ and $A_{R,t}$ and equipment efficiency $A_{E,t}$, each divided by its 1980 value. Panel (b) shows the manual share of output μ_t in percent.

4.5 Government

The baseline tax rates are time averages of the effective tax rates reported in Acemoglu et al. (2020): 25.59% on labor income over 1980–2018, 12.79% on net equipment income over 1981–2018, and 10.11% on net other-capital income, using the rate for nonresidential structures, over 1981–2018. The equipment tax rate averages the rates on equipment and software with their 1987 capital-cost weights, and we round it to 13% so that the status quo lies on the whole-percent grid of the policy search. Lump-sum transfers are zero in the baseline economy, and government expenditures satisfy the budget constraint. The consumption floor is 10% of initial average earnings. Social security benefits begin at age 66 and replace 38% of initial average earnings. Tax rates, the consumption floor, and social security benefits remain at their initial values.

Table 2: Calibration Parameters

Parameter	Description	Values	Source or target
Demographics			
J	Maximum age	61 (age 85)	Fixed
$j = 1$	Age at entry	25	Fixed
Human capital and wages			
σ_η	Productivity shock s.d.	0.25	PSID wage changes
Preferences and occupational choice			
β	Discount factor	0.97	Capital/output: 2.7
γ	Risk aversion	2.00	Fixed
B_s	Participation cost	1.43, 0.54	Participation at 65
θ_s	Preference shocks	0.17, 0.14	Routine entry shares
$\kappa_{s,o}, c^s_{oo'}$	Occupation/mobility costs	Table A.1	Shares and transitions
Production			
σ	Abstract vs. routine	0.60	Literature
ζ	Substitution across tasks	0.50	Humlum (2020)
χ	Comparative advantage	1.66	$\sigma_{LR} = 1.35$
b/χ	Slope of ψ_t above ι_0	0.95	Fixed
μ_{1980}	Manual share of output	0.04	Manual wage bill
α, s_E	Capital shares	0.16, 0.17	BLS (1987)
δ_E, δ_O	Depreciation rates	0.12, 0.03	BLS (1987)
Return and government			
r	Household return	4.21%	Acemoglu et al. (2020)
τ_I, τ_E, τ_O	Income tax rates	25.59%, 13%, 10.11%	Acemoglu et al. (2020)
$\underline{c}$	Consumption floor	0.10	Fixed
ss	Replacement rate	0.38	OECD (2017)
j_{ss}	Social security age	42 (age 66)	Fixed

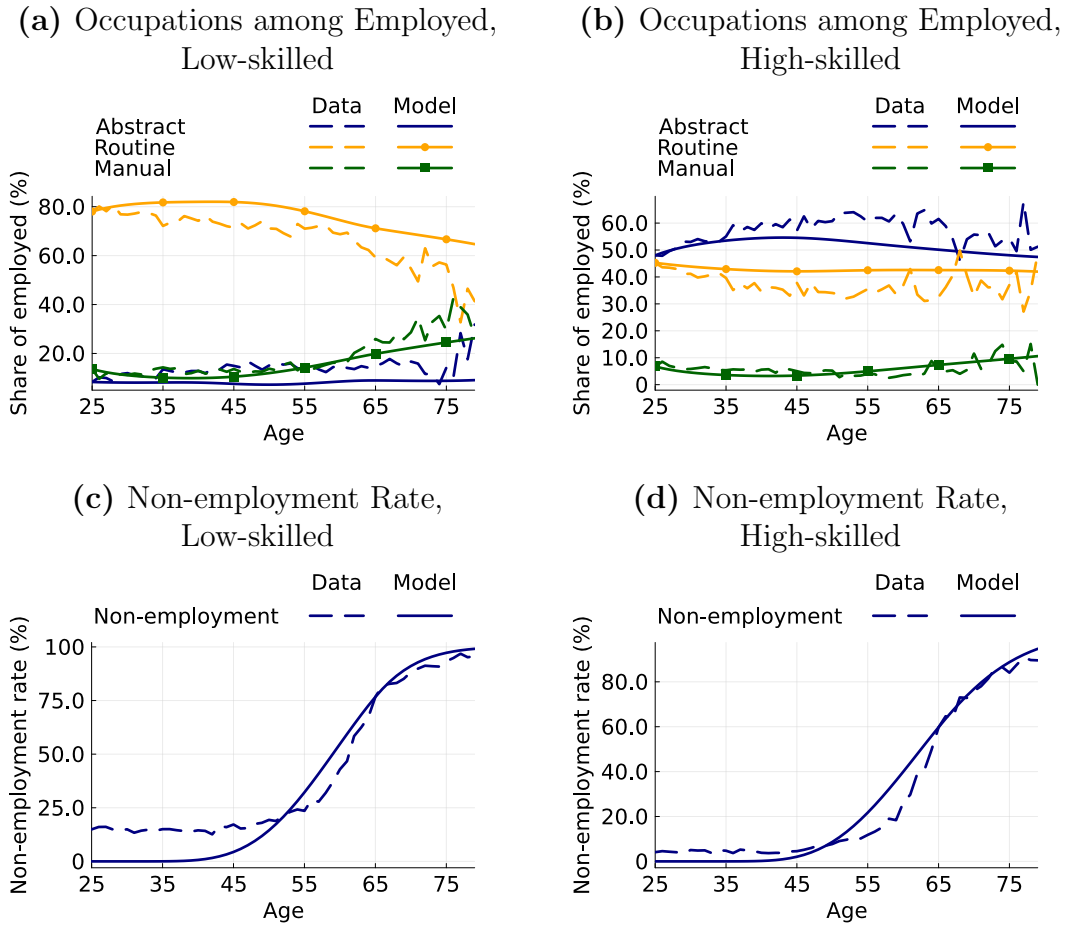
Notes: This table shows the model parameters. The two skill-specific values show low-skilled and high-skilled individuals, respectively. The initial economy uses 1983–1985 data. The parameter θ_s is the inverse scale of the preference shocks and is the same for entry and later occupational choice. The consumption floor is relative to initial average earnings.

4.6 Initial Economy

Figure 5 compares occupations and non-employment over the life cycle in the initial economy with the CPS in 1983–1985. The calibration targets occupation shares among employed individuals at age 25 and participation at age 65; the profiles at other ages are not targeted. The model captures the main life-cycle patterns in occupation shares and non-employment. Before age 50, non-employment is lower in the model than in the data, especially for the low-skilled, because all entrants in our model work and non-employment is a permanent exit from work. In the CPS, non-employment among low-skilled heads aged 25–29 mostly reflects

women outside the labor force for home duties: women are 28% of these heads but 65% of the non-employed, and home duties account for 52% of non-employment, above 31% for unemployment. The model includes neither family structure nor home production and abstracts from non-participation for these reasons.

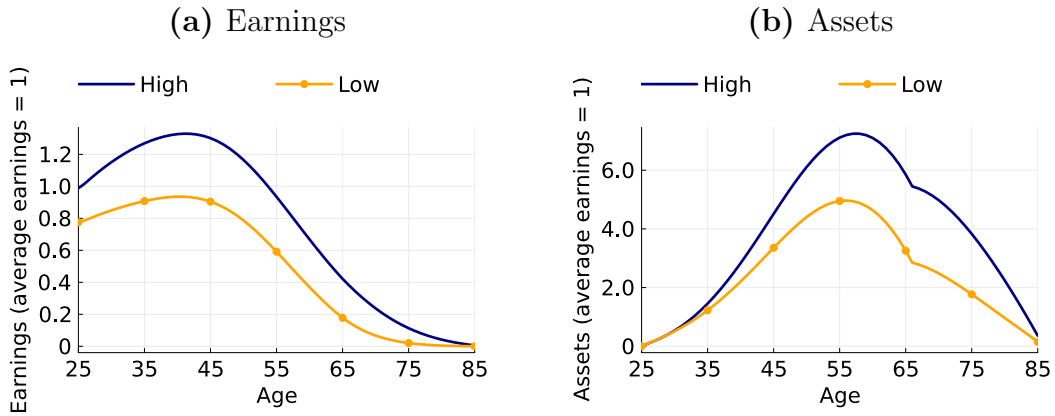
Figure 5: Occupations and Non-employment over the Life Cycle in the Initial Economy



Notes: This figure shows life-cycle profiles by skill group in the data and the initial model economy at ages 25–79. Panels (a) and (b) show the shares of abstract, routine, and manual occupations among employed individuals. Panels (c) and (d) show non-employment as a share of the population at each age. Dashed lines show household heads in the CPS in 1983–1985; solid lines show the model. Low-skilled individuals have at most a high school education; high-skilled individuals have at least some college education. The calibration targets occupation shares among employed individuals at age 25 and participation at age 65.

Figure 6 shows earnings and assets by age and skill in the initial economy. Earnings peak at about age 40 and then fall, mainly as individuals leave employment. Individuals enter with no assets, save for precautionary reasons and retirement, and decumulate assets after their late 50s; decumulation slows once social security benefits begin at age 66. High-skilled individuals have higher earnings at every age and accumulate more assets.

Figure 6: Earnings and Assets over the Life Cycle in the Initial Economy



Notes: This figure shows life-cycle profiles by skill group in the initial model economy at ages 25–85. Panel (a) shows average earnings and panel (b) average assets at the start of the year, over all individuals of each age and skill, including the non-employed. Both are in units of average earnings per worker in the initial economy, which the calibration sets to one.

5 Welfare Effects of Polarization

Since the early 1980s, employment has shifted from routine to abstract and manual occupations (Section 2), and wages have fallen in routine and manual occupations and risen in abstract occupations (Figure 3(d)). Who gained and who lost from this polarization? The effects may differ by skill, because in the initial economy 72.5% of low-skilled individuals aged 30–59 work in routine occupations, against 39.7% of the high-skilled. The welfare effects of polarization may also differ by age, because older workers have fewer working years left and more occupation-specific human capital to lose if they switch occupations. With the model calibrated in Section 4, this section measures these heterogeneous welfare effects for the generations alive in 1980 and for later generations.

We compare two cases with an economy in which technology remains at its 1980 level. In the constant-technology economy, the production technology represented by A_A, A_R, μ, and A_E remains at the 1980 level, and equipment productivity is one in every task, $\psi_t(i) \equiv 1$. The first case, automation only, replaces $\psi_t(i)$ with the profile in equation (13) and the path of (Υ_t, m_t) described in Section 4.4, and keeps the other technology at its 1980 level; we call it the automation economy. The second case, full polarization, is the baseline: it combines the same automation technology with the paths of $A_{A,t}$, $A_{R,t}$, μ_t, and $A_{E,t}$ that reproduce the occupational wages estimated from the CPS. Because it reproduces these wages, the baseline is also the status quo for the tax experiments in Section 6.

All economies have the same initial allocation in 1980, demographics, household preferences, return, and tax rates. Individuals learn the new paths at the

start of 1981. Both types of capital are inherited at that date, and the household return follows equation (21). In each economy, wages clear the labor markets and government expenditures balance the budget.[8]

We measure welfare by the consumption equivalent variation (CEV) of an alternative economy a relative to a benchmark economy b: the percentage change in consumption in the benchmark, applied in every state over the remaining lifetime, that makes an individual indifferent between the two economies. In this section, the benchmark is the constant-technology economy and the alternative is each case. Let $U^b(y,s) = U^b_c(y,s) + U^b_{\text{nc}}(y,s)$ denote expected utility from 1981 for skill s and entry cohort y in economy b. Here $U^b_c(y,s)$ is the consumption component and $U^b_{\text{nc}}(y,s)$ includes all nonconsumption utility: participation costs, occupation-specific and switching costs, and preference-shock utility. Utilities are averaged over individuals within the age and skill group, with survival and household discounting; individuals aged 85 in 1980 have no remaining life and are omitted. Explicitly,

$$U^b_c(y,s) \equiv \mathbb{E}_b\left[\sum_{n=0}^{T} \beta^n \frac{(c^b_n)^{1-\gamma}}{1-\gamma} \,\middle|\, y, s\right],$$

where $n = 0$ denotes the start of the cohort's welfare evaluation, T is the remaining lifetime, and the expectation includes survival and averaging within the cohort and skill group. Since only consumption is scaled, $\text{CEV}_{y,s}$ satisfies

$$(1 + \text{CEV}_{y,s})^{1-\gamma} U^b_c(y,s) + U^b_{\text{nc}}(y,s) = U^a_c(y,s) + U^a_{\text{nc}}(y,s). \tag{26}$$

Figures report $\text{CEV}_{y,s}$ in percent. A positive value means a gain in the alternative economy. For a cohort's aggregate CEV, we first compute average utility components across skills using population weights and then use equation (26). Figures show welfare effects for individuals alive in 1980 by their age and future generations by their year of entry at age 25.

5.1 Automation Only

We begin with automation alone. Automation is the technology that an automation tax targets, so its welfare effects show the distributional consequences of automation-driven polarization that such a tax might mitigate. The automation technology is identified from the wage effects estimated by Acemoglu and Restrepo (2022): by 2016, automation changes the wage index by -3.94 log points for the low-skilled and 12.12 for the high-skilled (Section 4.4). We first describe

[8] Transfers remain zero, and government spending absorbs changes in net revenue. The CEVs measure household welfare and assign no utility to government spending. In Section 6, spending follows the baseline path and lump-sum transfers adjust.

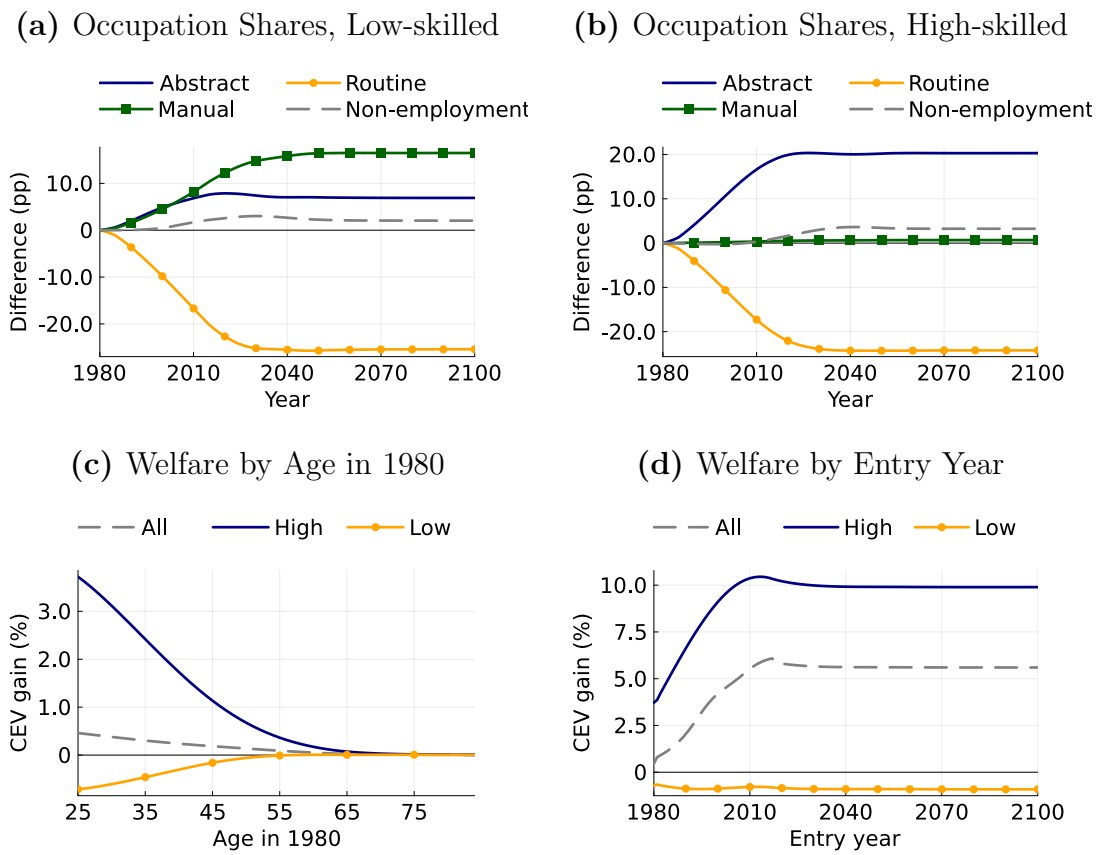
how these effects arise across occupations and how occupational choices respond, and then measure welfare effects across skills and generations.

Automation raises the productivity of equipment in routine tasks, so equipment performs more of them. The routine wage falls; the abstract wage rises, because abstract labor complements routine tasks; and the manual wage rises with output. Relative to constant technology, log wages in 2016 change by +27.1 in abstract, −9.4 in routine, and +14.4 in manual occupations, measured as log changes multiplied by 100. Panels (a) and (b) of Figure 7 show how occupational choices respond, as differences in occupation shares from those under constant technology. Among the low-skilled, employment shifts from routine mainly to manual occupations: in 2018, among individuals aged 30–59, their routine share changes by −21.8 percentage points and their manual share by +11.5 points. Among the high-skilled, the abstract share changes by +19.5 points. Automation also raises non-employment by 2.4 points for the low-skilled and 1.3 points for the high-skilled. The shares describe repeated cross-sections, including new entrants and surviving incumbents, so a declining routine share need not represent a transition out of routine work by the same people; at the end of this subsection, we follow people working in routine occupations in 1980.

Panels (c) and (d) of Figure 7 show welfare gains relative to constant technology. For entrants in 1980, the aggregate CEV is 0.46%: −0.71% for the low-skilled and 3.72% for the high-skilled. Among individuals aged 40 in 1980, the CEV is −0.30% for the low-skilled and 1.73% for the high-skilled. At age 65, it is 0.01% and 0.07%, respectively. Individuals near retirement in 1980 are hardly affected, because the wage effects of automation build up from 1980 to 2016, few of their working years remain, and social security benefits do not change with wages; for younger individuals and later entrants, more of the working life is exposed to the new wages, so both the losses of the low-skilled and the gains of the high-skilled grow.

For entrants in 2018, the CEV is −0.83% for the low-skilled and 10.29% for the high-skilled; for entrants in 2100, it is −0.91% and 9.89%. These entrants face the full wage effects over their working lives: the welfare loss of the low-skilled is small relative to the −3.94 log-point change in their wage index, whereas the welfare gain of the high-skilled is close to the 12.12 log-point change in theirs; Section 5.3 explains the small loss of the low-skilled. The aggregate CEV of entrants in 2100 is 5.60%, as the high-skilled share of entrants is larger than in 1980. The aggregate CEV is positive for all cohorts, from individuals aged 84 in 1980 to entrants in 2100, although automation raises non-employment.

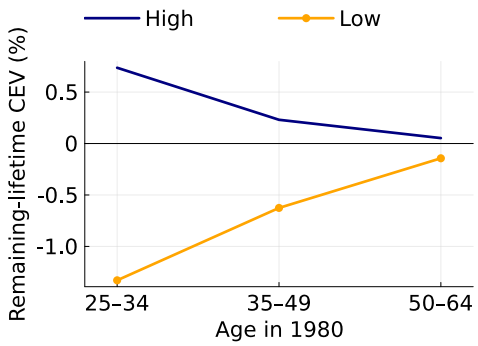
Figure 7: Occupations and Welfare: Automation Only



Notes: This figure shows the automation economy, which adds the automation technology identified from the wage effects in Acemoglu and Restrepo (2022) to constant 1980 technology. Panels (a) and (b) show the differences in occupation shares among individuals aged 30–59 from the economy with technology fixed at its 1980 level, from 1980 to 2100, in percentage points; shares include non-employment. Panels (c) and (d) show welfare gains relative to that economy, for ages 25–84 in 1980 and for entry years 1980–2100. All denotes aggregate CEV, computed from population-weighted utility across skills; High and Low denote high-skilled and low-skilled individuals. Low-skilled individuals have at most a high school education; high-skilled individuals have at least some college education. Gains are in percentage terms and count utility from 1981. A positive value means a welfare gain in the automation economy.

Figure 8 follows people working in routine occupations in 1980. Groups are defined by occupation, skill, age, and beginning-of-1980 assets, and membership stays fixed across economies. We record single ages and pool them into ages 25–34, 35–49, and 50–64. Welfare counts remaining life from 1981, discounted to 1980, per initial member. The CEV of automation is −1.33% for low-skilled routine workers aged 25–34 and −0.14% for those aged 50–64, and 0.74% and 0.05% for the high-skilled of the same age groups. Conditioning on routine employment before the shock therefore preserves the age gradient of Figure 7(c). Older workers spend fewer remaining working years under the new wage path, limiting both losses and gains. A common initial occupation still allows different exposure through skill-specific earnings profiles and subsequent choices.

Figure 8: Remaining-Lifetime Welfare of Initial Routine Workers: Automation Only



Notes: This figure shows full CEV in the automation economy relative to constant technology for groups working in routine occupations in 1980, with fixed membership across economies. High and Low denote high-skilled and low-skilled routine workers. Age refers to 1980; utility begins in 1981, includes survival, and is averaged per initial member before computing CEV. Asset groups are pooled using their initial masses.

5.2 Full Polarization

Automation alone does not reproduce the wages observed in the data. In the CPS, wages change from 1980 to 2018 by +8.2% in abstract, −10.4% in routine, and −2.7% in manual occupations, whereas in the automation economy the manual wage rises. Full polarization in the baseline economy reproduces these wages, so its welfare effects include the changes in occupational wages from sources other than automation. Relative to the case of constant technology, log wages in 2016 change by +18.8 in abstract, −15.0 in routine, and −7.8 in manual occupations: compared with automation only, the abstract wage rises less, and the routine and manual wages are lower.

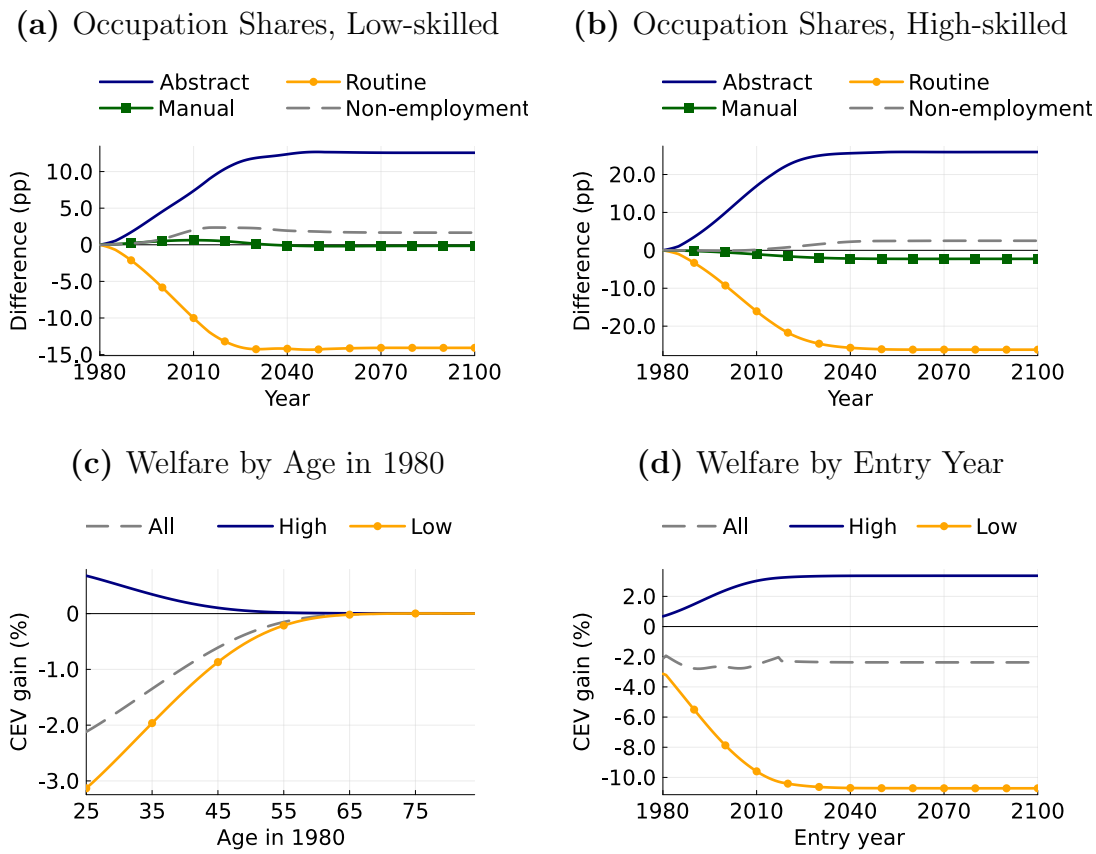
Panels (a) and (b) of Figure 9 show the differences in occupation shares from constant technology. Unlike under automation only, employment of the low-skilled shifts from routine mainly to abstract occupations: in 2018, among individuals aged 30–59, their routine share changes by −12.7 percentage points, their abstract share by +9.8 points, and their manual share by +0.5 points. Among the high-skilled, the abstract share changes by +21.6 points. Non-employment changes by +2.4 points for the low-skilled and +0.7 points for the high-skilled.

The effect on non-employment changes after wages reach their final values, because individuals who entered under earlier wages have different assets and human capital from later entrants. For the low-skilled, the difference in non-employment from constant technology is +2.4 points in 2018 and +1.7 points in 2100.

Panels (c) and (d) of Figure 9 show welfare gains relative to constant technology. For entrants in 1980, the CEV is −3.13% for the low-skilled and 0.68% for

the high-skilled; for entrants in 2018, it is −10.33% and 3.24%; and for entrants in 2100, −10.72% and 3.37%. Relative to automation only, the low-skilled lose much more and the high-skilled gain less, in line with these wage effects. The two cases are separate equilibrium experiments; their CEVs do not form an additive decomposition of historical welfare changes. Each comparison includes the household responses and fiscal adjustment of its own equilibrium.

Figure 9: Occupations and Welfare: Full Polarization



Notes: This figure shows full polarization (the baseline), which combines the recovered automation technology with paths of occupational productivity and equipment efficiency that reproduce CPS wages. Panels (a) and (b) show the differences in occupation shares among individuals aged 30–59 from the economy with technology fixed at its 1980 level, from 1980 to 2100, in percentage points; shares include non-employment. Panels (c) and (d) show welfare gains relative to that economy, for ages 25–84 in 1980 and for entry years 1980–2100. All denotes aggregate CEV, computed from population-weighted utility across skills; High and Low denote high-skilled and low-skilled individuals. Low-skilled individuals have at most a high school education; high-skilled individuals have at least some college education. Gains are in percentage terms and count utility from 1981. A positive value means a welfare gain in full polarization.

5.3 Sources of Welfare Changes

Table 3 decomposes the welfare changes of entrants in 2018 in the two cases. For entry cohort y and skill s, let $U_k^e(y, s)$ denote expected discounted lifetime utility from component k in economy e, accounting for survival and averaging

across individuals within each group. Here a denotes the alternative economy, b the constant-technology benchmark, and c the consumption component. For each utility component k (consumption, participation costs, occupation-specific costs, switching costs, and preference-shock utility), we measure the change as a percentage of benchmark consumption, using a first-order approximation:

$$\frac{100\Delta U_k(y,s)}{(1-\gamma)U_c^b(y,s)}, \qquad \Delta U_k(y,s) = U_k^a(y,s) - U_k^b(y,s). \tag{27}$$

The denominator $(1-\gamma)U_c^b(y,s)$ is the marginal utility gain from a unit proportional increase in benchmark consumption for cohort y and skill s. Panel A reports consumption, participation, and the net value of occupational choice, which is the sum of the last three components; Table A.2 in the Appendix reports them separately. We combine them because individuals who choose occupations with higher occupation-specific or switching costs are selected on favorable preference shocks for those occupations. A shift in occupational choices therefore moves these costs and preference-shock utility in opposite directions. Panel B reports exact CEVs. Equation (26) can be written as $1+\text{CEV}_{y,s} = (1+\Delta U(y,s)/U_c^b(y,s))^{1/(1-\gamma)}$, where $\Delta U(y,s) = \sum_k \Delta U_k(y,s)$; the sum in Panel A approximates $100 \times \text{CEV}_{y,s}$ to first order. The consumption-only CEV applies the same formula to the consumption component alone and excludes the other utility changes. It is a diagnostic of the consumption channel, not an alternative social objective.

For low-skilled entrants, automation has a consumption-only CEV of -2.88%, compared with a full CEV of -0.83%. The smaller full loss reflects lower participation costs and a net gain from occupational choice of 0.94% of benchmark consumption. This small net gain combines large changes of opposite sign: occupation-specific and switching costs rise, and preference-shock utility rises by 12.83% (Table A.2). Preference-shock utility is utility from the chosen occupations, with no corresponding addition to resources. For high-skilled entrants, consumption accounts for most of the automation gain: consumption-only CEV is 9.19%, compared with a full CEV of 10.29%.

Nonconsumption terms have a different net effect in full polarization. Consumption-only CEV is -8.23% for the low-skilled and 5.41% for the high-skilled; including all utility terms increases the low-skilled loss and reduces the high-skilled gain. In both skill groups, the decline in the net value of occupational choice exceeds the reduction in participation costs. For the high-skilled, lower switching costs are more than offset by lower preference-shock utility and higher occupation-specific costs (Table A.2).

Table 3: Welfare Components of the 2018 Entry Cohort

	Automation only		Full polarization	
	Low	High	Low	High
Panel A. Utility change by component				
Consumption	−2.97	8.42	−8.97	5.13
Participation	1.19	1.08	0.66	0.74
Occupational choice (net)	0.94	−0.16	−3.22	−2.73
Sum	−0.83	9.33	−11.53	3.14
Panel B. Consumption equivalent variation				
Consumption only	−2.88	9.19	−8.23	5.41
All components	−0.83	10.29	−10.33	3.24

Notes: This table shows welfare changes for 2018 entrants by skill, with automation only and with full polarization, each relative to constant technology. Panel A reports $100\Delta U_k(y,s)/[(1-\gamma)U_c^b(y,s)]$, where $\Delta U_k(y,s) = U_k^a(y,s) - U_k^b(y,s)$ is the utility change in component k for entry cohort y and skill s, a denotes the alternative economy, b the constant-technology benchmark, and $U_c^b(y,s)$ its consumption utility; the sum is over k. Occupational choice (net) is the sum of occupation-specific costs, switching costs, and preference-shock utility. Positive entries raise welfare, including when they represent reduced costs. Panel B reports exact CEVs in percent. The "Consumption only" row scales consumption utility and excludes the other components; the "All components" row reports the CEV shown in the figures. The sum in Panel A approximates $100 \times \text{CEV}_{y,s}$ to first order. Survival, discounting, and entry costs follow the lifetime welfare measure.

6 Automation Tax

Section 5 shows that automation raises average welfare in every generation but lowers the welfare of low-skilled entrants. Several studies argue for taxing automation, because adversely affected workers can adjust only gradually or because capital income is taxed at lower rates than labor income in the U.S. (Acemoglu et al. 2020; Guerreiro et al. 2022; Beraja and Zorzi 2025). This section quantitatively investigates how such a tax can mitigate the losses of the low-skilled, and who bears its cost. The average welfare gain from automation does not answer this question, because a change in the tax redistributes across skills and generations.

We change the tax rate on net equipment income, which we refer to as the automation tax, from its status quo level of 13% to a new constant rate starting in 1981 and maintained thereafter. To account for the fiscal effects of this policy change, we assume that the government rebates any additional tax revenue through uniform lump-sum transfers. The labor income tax remains at 25.59%, the tax on other capital remains at 10.11%, and government expenditures follow the baseline path. Firms take the baseline paths of $A_{A,t}$, $A_{R,t}$, μ_t, $A_{E,t}$, and the automation technology as given, and wages clear the labor markets. Demo-

graphics and benefit rules are the same as in the baseline. The tax applies to all equipment rather than only to newly automated tasks, and we do not consider targeted subsidies, age-dependent transfers, or time-varying tax paths.

6.1 Social Welfare

The optimal automation tax maximizes welfare across individuals alive in 1980 and future entrants. Following Heathcote et al. (2020), we use equal weights for individuals within a cohort and the household discount factor for future cohorts. Let $t = 0$ denote 1980 and Γ denote a policy. Social welfare is

$$\mathcal{W}(\Gamma) = (1-\beta)\left[\sum_{j=1}^{J}\sum_{s} N_{j,s,0}U^{\text{old}}_{j,s,0}(\Gamma) + \sum_{t=1}^{\infty}\beta^{t}\sum_{s} N^{\text{ent}}_{s,t}U^{\text{ent}}_{s,t}(\Gamma)\right]. \tag{28}$$

$U^{\text{old}}_{j,s,0}$ is remaining expected utility per person, discounted to 1980, and $U^{\text{ent}}_{s,t}$ is expected lifetime utility per entrant, discounted to the date of entry. Both account for survival risks. For individuals alive in 1980, $U^{\text{old}}_{j,s,0}$ includes utility from 1981, as in Section 5.

We compute welfare effects for future entrants through 2200 and use the final stationary economy thereafter. If $\mathcal{T} = 220$ denotes the number of years from 1980 to 2200, the sum for entrants after $\mathcal{T}$ in equation (28) is

$$\frac{\beta^{\mathcal{T}+1}}{1-\beta}\sum_{s} N^{\text{ent}}_{s,*}U^{\text{ent}}_{s,*}(\Gamma), \tag{29}$$

where $*$ denotes the final stationary economy. Aggregate welfare gains in CEV are computed from consumption and nonconsumption utility components using equation (26) and the weights in equation (28).

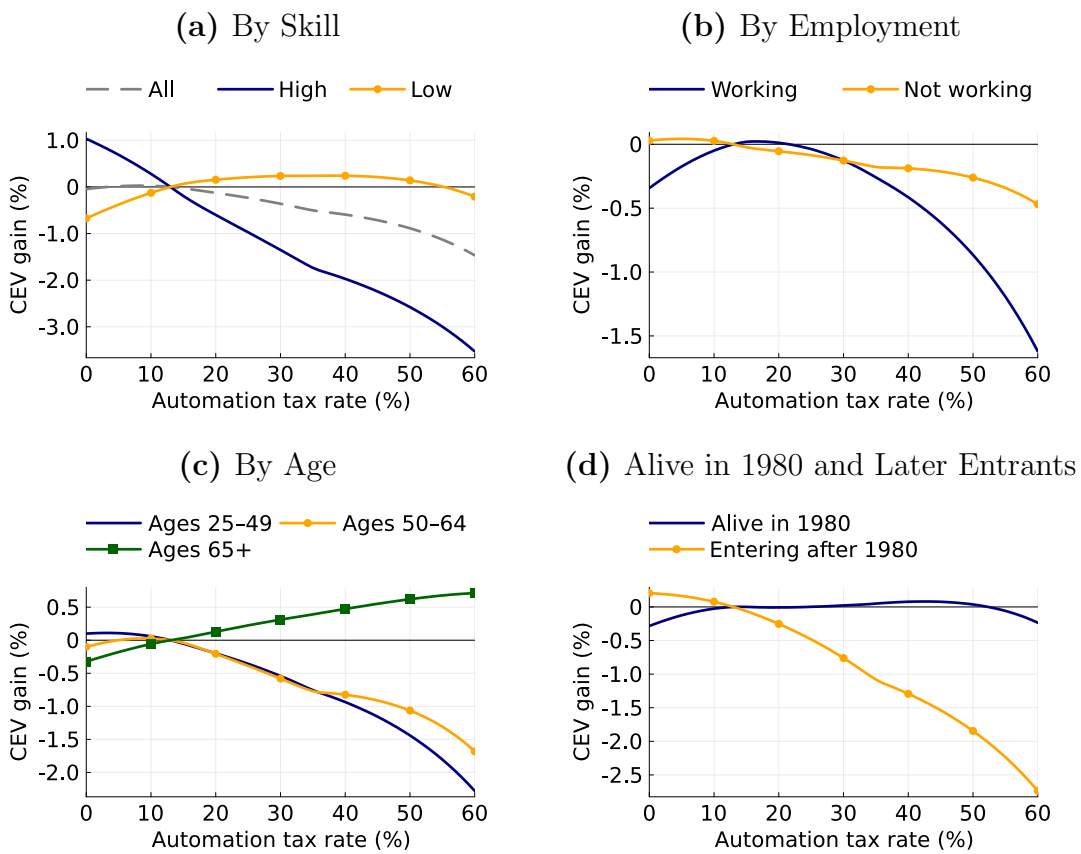
The tax rate trades off equipment costs, occupational wages, transfers, and the return on inherited capital. A lower automation tax reduces the rental cost of equipment and changes firms' assignment of tasks; after capital adjusts, abstract and manual wages rise, and the routine wage falls. The transfer must finance the revenue change while spending stays fixed. On impact, equipment and other capital are inherited, so their payout and the household asset return also respond. These effects differ across skills, assets, and remaining working lives, and the optimal tax reflects their combined discounted utility effects.

6.2 Welfare by Tax Rate

We solve the transition at every whole-percent tax rate from 0% to 60%. Figure 10 shows welfare gains relative to the baseline. Social welfare is maximized at a

tax rate of 8%, below the status quo of 13%, with a gain of 0.03% in consumption-equivalent terms.[9] The welfare gain is nearly flat around the optimum: tax rates between 7% and 10% yield gains within 0.005 percentage points of the maximum. When social welfare is evaluated separately for each skill group, the optimal tax rates differ substantially. The low-skilled prefer a rate of 38%, with a welfare gain of 0.24%, whereas the high-skilled prefer a rate of 0%, with a gain of 1.03%.

Figure 10: Welfare Effects by Automation Tax Rate: Lump-sum Transfers



Notes: This figure shows CEV gains relative to the baseline, in which the automation tax is 13%, for constant automation tax rates from 0% to 60% imposed from 1981. The change in revenue is paid as a uniform lump sum to everyone alive; the labor income tax stays at 25.59% and government expenditures follow the baseline path. Panel (a) shows social welfare (All) and the same welfare for high-skilled (High) and low-skilled (Low) individuals. Panels (b) and (c) group each year's population by its state in that year, so an individual moves between groups over the life cycle: by employment in panel (b) and by age in panel (c). Each group's utility in each year from 1981 is discounted to 1980 as in social welfare, and the gain compares utility per member and year, since employment responds to the tax. Panel (d) splits social welfare into individuals alive in 1980 and later entrants. Gains are in percentage terms; the status quo reproduces baseline welfare up to numerical precision.

[9] With entrants computed through 2240, the optimal rate is 8%.

Table 4: Economies by Automation Tax Rate: Lump-sum Transfers

		Automation tax rate (%); status quo: 13%				
		0	10	20	30	40
CEV gain (%)						
All		−0.05	0.02	−0.13	−0.36	−0.59
High		1.03	0.28	−0.60	−1.35	−1.97
Low		−0.67	−0.13	0.15	0.24	0.24
Working		−0.34	−0.05	0.01	−0.13	−0.41
Not working		0.03	0.03	−0.05	−0.13	−0.19
Ages 25–49		0.10	0.06	−0.20	−0.54	−0.94
Ages 50–64		−0.10	0.02	−0.21	−0.58	−0.82
Ages 65+		−0.32	−0.06	0.13	0.31	0.47
Alive in 1980		−0.28	−0.03	−0.01	0.02	0.08
Entering after 1980		0.21	0.08	−0.25	−0.76	−1.29
Output (%)	2018	7.2	1.7	−4.2	−10.5	−12.7
	2300	6.2	1.5	−3.6	−9.2	−12.9
High wage (%)	2018	1.2	0.3	−0.7	−1.8	−2.9
	2300	0.9	0.2	−0.6	−1.5	−2.4
Low wage (%)	2018	−0.9	−0.2	0.6	1.5	1.2
	2300	−1.1	−0.3	0.7	1.8	2.1
Routine wage (%)	2018	−1.6	−0.4	1.0	2.7	2.7
	2300	−1.8	−0.4	1.1	2.9	3.7
Routine share (pp)	2018	−3.7	−0.9	2.1	5.1	5.9
	2300	−3.5	−0.8	2.1	5.3	7.4
Non-employment share (pp)	2018	0.8	0.2	−0.4	−0.8	−0.9
	2300	0.7	0.2	−0.5	−1.2	−1.7
Automation share (pp)	2018	6.5	1.5	−3.7	−8.6	−10.0
	2300	6.5	1.6	−3.7	−9.2	−12.5

Notes: This table shows the economy under constant automation tax rates imposed from 1981 relative to the status quo, in which the tax is 13%; CEV gains compare welfare under each policy with welfare under the status quo; annual outcomes are differences from the status quo in the same year. The change in revenue is paid as a uniform lump sum to everyone alive; the labor income tax stays at 25.59% and government expenditures follow the baseline path. CEV gains are for social welfare (All), which weights individuals alive in 1980 and future entrants discounted to 1980, and the same welfare for high-skilled (High) and low-skilled (Low) individuals; each year's population grouped by employment and by age in that year; and individuals alive in 1980 and later entrants. Output and wages are percentage differences: output, the high- and low-skill wages, which are wage indices that weight log occupational wages by the 1980 earnings shares of the high-skilled and the low-skilled, and the routine wage. Shares are differences in percentage points (pp): the share of routine occupations among employed individuals aged 30–59, the share of individuals aged 30–59 who are not employed, and the share of automated tasks. 2300 is the final stationary economy.

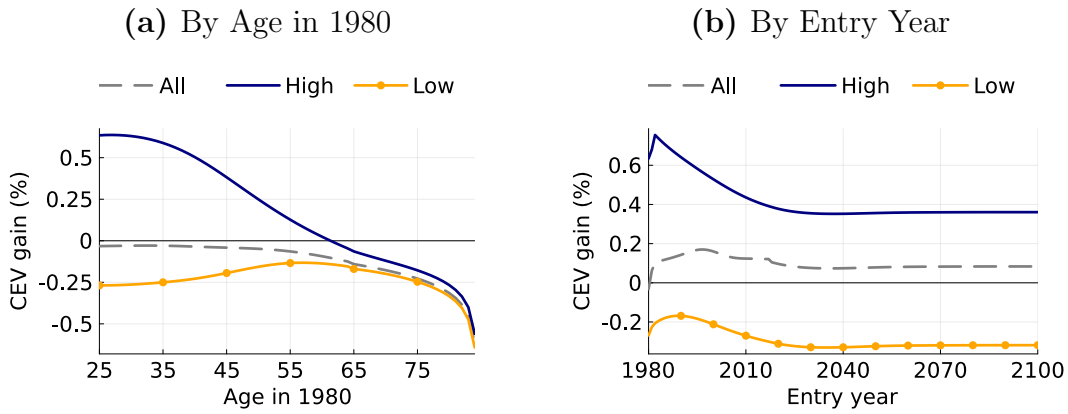
Panels (b) and (c) group each year's population by its state in that year, so an individual moves between groups over the life cycle. Individuals aged 65 and above gain from rates above the status quo, by 0.71% at 60%. The transfers benefit this

group, which has lower employment rates than younger individuals. Individuals aged 25–49 have a CEV of −2.28% at 60%. Panel (d) splits social welfare into the population alive in 1980, which is best off at 43% with a gain of 0.08%, and later entrants, who are best off at 0% with a gain of 0.21%. The population alive in 1980 owns the inherited assets and has shorter exposure to subsequent wages, whereas entrants face those wages over their full careers. Because social welfare pools utility across people and dates before converting it to consumption units, the optimal rate cannot be recovered by averaging the rates preferred by groups. Table 4 reports these gains at selected tax rates together with output, wages, and the shares of routine occupations, non-employment, and automated tasks.

6.3 The Preferred Tax Rates

The Optimal Tax. Because the optimal rate of 8% is below the status quo, transfers are negative: lump-sum taxes finance the tax cut. As a share of average consumption in each year, transfers are −0.40% in 1981, −0.14% in 2018, and −0.13% in 2100. Once equipment adjusts, the abstract, manual, and routine wages in 2018 differ from the baseline by 1.27%, 0.62%, and −0.66%, respectively. Figure 11 shows welfare gains under the optimal tax relative to the baseline. For entrants in 1980, the aggregate CEV is −0.03%: −0.27% for the low-skilled and 0.63% for the high-skilled. At age 65 in 1980, it is −0.17% for the low-skilled and −0.06% for the high-skilled. For entrants in 2100, it is −0.32% for the low-skilled and 0.36% for the high-skilled, with an aggregate CEV of 0.08%.

Figure 11: Welfare Effects of Optimal Automation Tax



Notes: This figure shows CEV gains under the 8% optimal automation tax relative to the baseline, in which the automation tax is 13%. All denotes aggregate CEV, computed from population-weighted utility across skills. High and Low denote high-skilled and low-skilled individuals. Panel (a) shows ages 25–84 in 1980; panel (b) shows entry years 1980–2100. Gains are in percentage terms and count utility from 1981, as in social welfare. A positive value means a welfare gain under the tax. The labor income tax is 25.59% and government expenditures follow the baseline path.

Utility Channels of the Policy Tradeoff. Table 5 decomposes welfare changes at the aggregate-optimal rate of 8% and at the low-skill-optimal rate of 38% into consumption and nonconsumption utility components, expressed in benchmark-consumption units, and reports the full CEV. Nonconsumption utility combines participation, occupation-specific, and switching costs with preference-shock utility. At the aggregate-optimal rate of 8%, consumption utility rises in the aggregate and for the high-skilled, but falls for the low-skilled. Nonconsumption utility falls slightly in every group, so the small full welfare gain reflects the aggregate increase in consumption utility.

At the low-skill-optimal rate of 38%, low-skilled consumption utility rises and nonconsumption utility changes little. The resulting gain of 0.24% comes with losses for the high-skilled and later entrants. Among people alive in 1980, the full CEV is positive even though their consumption component is negative, because their nonconsumption utility rises. The contrast with later entrants highlights the importance of transfer payments, remaining working years, and utility changes associated with occupational choices in determining the welfare effects of the automation tax.

Table 5: Utility Components at the Two Preferred Tax Rates

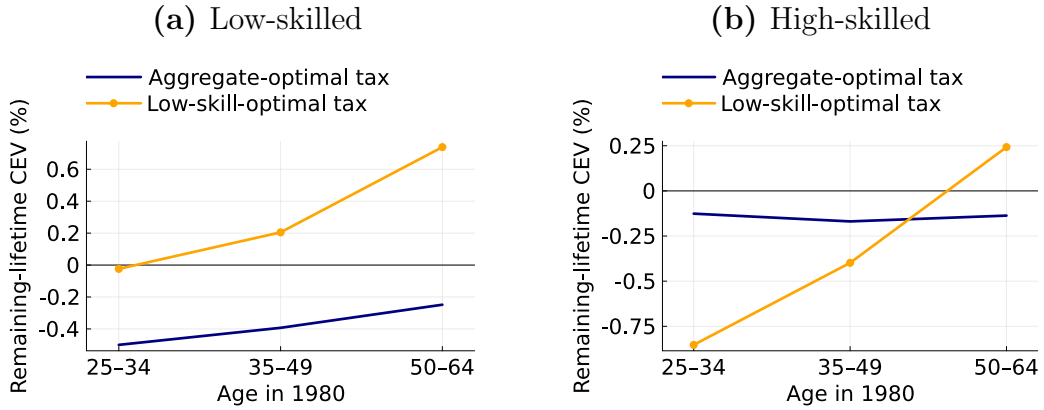
	Aggregate optimum: 8%			Low-skill optimum: 38%		
	Cons.	Noncons.	CEV	Cons.	Noncons.	CEV
All	0.09	−0.06	0.03	−0.66	0.10	−0.56
Low	−0.16	−0.06	−0.22	0.18	0.06	0.24
High	0.51	−0.06	0.45	−2.10	0.19	−1.88
Alive in 1980	−0.02	−0.04	−0.06	−0.22	0.29	0.07
Later entrants	0.20	−0.08	0.12	−1.13	−0.09	−1.21

Notes: This table shows utility changes relative to the baseline at the aggregate-optimal automation tax rate of 8% and the low-skill-optimal rate of 38%. Cons. (consumption) and Noncons. (nonconsumption) are additive utility changes in benchmark-consumption units; nonconsumption combines participation, occupation-specific, and switching costs with preference-shock utility. CEV is the exact full CEV and includes both components. All, Low, and High use the social-welfare weights; Alive in 1980 and Later entrants split that criterion into people alive in 1980 and later entrants. Utilities are pooled before conversion to consumption units.

Initial Routine Workers. Figure 12 shows the welfare effects of the aggregate-optimal and low-skill-optimal tax rates for the initial routine workers examined in Figure 8, relative to the baseline. At the aggregate-optimal rate of 8%, all groups of initial routine workers lose: the CEV is −0.50% for low-skilled workers aged 25–34 and −0.25% for those aged 50–64, and −0.13% and −0.14% for the high-skilled. At the low-skill-optimal rate of 38%, welfare gains rise with age for both skill groups: from −0.02% to 0.74% for the low-skilled and from −0.85%

to 0.24% for the high-skilled. Older workers benefit more from the higher tax because they have fewer remaining working years exposed to the resulting wage changes and receive lump-sum transfers during retirement.

Figure 12: Remaining-Lifetime Welfare of Initial Routine Workers at the Two Preferred Tax Rates



Notes: This figure shows full CEV for groups working in routine occupations in 1980, with fixed membership across economies, under the aggregate-optimal tax of 8% and the low-skill-optimal tax of 38%, each compared with the baseline. Age refers to 1980; utility begins in 1981, includes survival, and is averaged per initial member before computing CEV. Asset groups are pooled using their initial masses.

6.4 Robustness

Table 6 reports the optimal rate and the associated welfare gains under three alternative specifications: transfer payments restricted to workers, labor market rents as in Acemoglu et al. (2020), and manual labor included in the CES aggregate with abstract labor and routine tasks. The optimal tax rate ranges from 8% to 10%, a range over which social welfare in the main specification is nearly flat, varying by less than 0.005 percentage points from its maximum. Across all specifications, the high-skilled gain and the low-skilled lose at the optimal tax rate.

Table 6: Optimal Automation Tax under Alternative Specifications

	Optimal	CEV gain (%)		
	rate (%)	All	High	Low
Main specification	8	0.03	0.45	−0.22
Transfers to workers	9	0.02	0.35	−0.17
Labor market rents, small ($\varrho = 7.5\%$)	9	0.02	0.36	−0.18
Labor market rents, large ($\varrho = 15\%$)	10	0.01	0.27	−0.13
Manual labor in the CES aggregate	8	0.03	0.45	−0.21

Notes: This table shows, for each specification, the constant automation tax rate from 1981 that maximizes social welfare, and CEV gains at that rate relative to the status quo of the same specification, in which the tax is 13%. Rates are whole percents from 0% to 60% in the main specification, and whole percents from 0% to 20% and 5-point steps to 60% in the others; social welfare falls at every rate above the optimum, so the optimal rate is exact to the whole percent. All denotes social welfare; High and Low denote the same welfare for high-skilled and low-skilled individuals. In the main specification, the change in revenue is paid as a uniform lump sum to everyone alive. With transfers to workers, it is paid only to individuals who work in that year. With labor market rents, each year of work adds the rent with wedge ϱ at the allocations of the main specification. With manual labor in the CES aggregate, manual labor enters X_t with abstract labor and routine tasks, and the automation technology, the baseline paths and the tax experiment are solved again.

Transfers to Workers. The change in tax revenue is redistributed each year through lump-sum transfers to workers only. Unlike in the baseline, these transfers affect the incentive to work: when the tax rate is below the status quo, transfers are negative and effectively impose an additional tax on employment. Households anticipate this rule, and the equilibrium is solved to account for changes in employment, saving, and wages. Social welfare is maximized at a tax rate of 9%, compared with 8% in the main specification. At the optimal rate, both the welfare gain for the high-skilled and the loss for the low-skilled are smaller than in the main specification. Figure A.1 and Table A.3 in the appendix report welfare effects by group and the economic outcomes at selected tax rates.

Labor Market Rents. Following Acemoglu et al. (2020), we assume that the wage exceeds the opportunity cost of work by a wedge ϱ. We keep the allocations from the main specification and add an employment rent of $\varrho(1-\tau_{I,t})y$ on earnings y for each year of work, valued at the 1980 marginal utility of consumption of workers with the same skills. We consider $\varrho = 7.5\%$ and $\varrho = 15\%$. Social welfare is maximized at tax rates of 9% and 10%, respectively. At both rates, the high-skilled gain and the low-skilled lose. This calculation changes the welfare value assigned to employment while holding allocations fixed; household decisions, wages, and tax revenue remain unchanged. Thus, the changes in the optimal tax rate reflect the sensitivity of welfare evaluations to employment rents, rather

than the equilibrium effects of introducing labor-market frictions. Figure A.2 and Table A.4 in the appendix report welfare effects by group.

Manual Labor in the CES Aggregate. Manual labor, abstract labor, and routine tasks enter the composite with a common elasticity of substitution, σ,

$$X_t = \left[A_{A,t} L_{A,t}^{\frac{\sigma-1}{\sigma}} + A_{R,t} Z_t^{\frac{\sigma-1}{\sigma}} + A_{M,t} L_{M,t}^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}, \qquad Y_t = \left(\frac{K_{O,t}}{\alpha}\right)^{\alpha} \left(\frac{X_t}{1-\alpha}\right)^{1-\alpha},$$

so that $A_{M,t}$ replaces μ_t. The household parameters are unchanged, because the calibration takes wages as given and initial output does not depend on how manual labor enters production. We recalibrate the production coefficients and automation technology and recompute the baseline and tax experiments. The baseline reproduces the same wages and allocation. In the automation economy, A_A and A_M remain at their 1980 values, so $w_A/w_M = (A_A/A_M)(L_A/L_M)^{-1/\sigma}$: holding labor inputs fixed, automation raises the manual wage as much as the abstract wage, and the abstract wage rises more only if manual employment grows faster than abstract employment. Matching the wage effects in Acemoglu and Restrepo (2022), the manual wage rises by 16.3 log points from 1980 to 2016 relative to constant technology, compared with 14.4 with the Cobb-Douglas weight (Section 5). The corresponding changes are 27.2 versus 27.1 log points for the abstract wage and -9.6 versus -9.4 log points for the routine wage. With the larger gain in the manual wage, the CEV of automation for low-skilled entrants in 2018 is -0.30%, against -0.83% with the Cobb-Douglas weight. The optimal tax remains at 8%, and the welfare gains by skill at this rate are similar to those of the main specification. Figure A.3 and Table A.5 in the appendix report welfare effects by group and the economy at selected rates.

7 Conclusion

We study the welfare effects of labor market polarization and of an automation tax in a life-cycle model with task-based production and occupational mobility. With the automation technology identified from the wage effects in Acemoglu and Restrepo (2022), automation raises average welfare in every generation, but its effects differ substantially across skills and generations. Low-skilled entrants lose, and high-skilled entrants gain. Individuals near retirement in 1980 are hardly affected, whereas younger workers and later entrants experience larger welfare effects because they face the new wages over more of their working lives.

With the occupational wages estimated from the CPS, which also reflect changes other than automation, the welfare losses of the low-skilled are much

larger. Unlike under automation alone, manual wages decline, while routine wages fall further, worsening the labor market outcomes of low-skilled workers. For low-skilled entrants in 2018, the consumption-equivalent welfare change is -10.33% under full polarization, compared with -0.83% under automation alone.

Raising the automation tax can benefit the low-skilled through higher routine wages and the generations alive in 1980 through lump-sum transfers, but at the expense of the high-skilled and future generations. Social welfare instead favors a small cut in the tax, from 13% to 8%. Once equipment adjusts, the lower tax raises the abstract wage and lowers the routine wage, benefiting high-skilled entrants while hurting low-skilled entrants. The aggregate welfare gain is only 0.03% in consumption-equivalent terms, small relative to the redistribution across skills and generations. This result is robust to alternative specifications in which transfers are restricted to workers, employment carries rents, or manual labor enters the CES aggregate.

We consider constant tax rates with uniform lump-sum transfers. Whether targeted subsidies, age-dependent transfers, or time-varying tax policies can mitigate the distributional consequences of automation at a lower welfare cost remains an important question for future research.

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Appendix: Additional Figures and Tables

Table A.1: Calibrated Mobility and Occupation Costs

(a) Low-skilled

	Abstract	Routine	Manual
Abstract	–	29.38	31.58
Routine	19.78	–	23.35
Manual	25.73	24.96	–
Occupation-specific	1.22	0.00	0.13

(b) High-skilled

	Abstract	Routine	Manual
Abstract	–	28.76	35.13
Routine	24.74	–	28.09
Manual	34.68	31.94	–
Occupation-specific	0.11	0.00	0.57

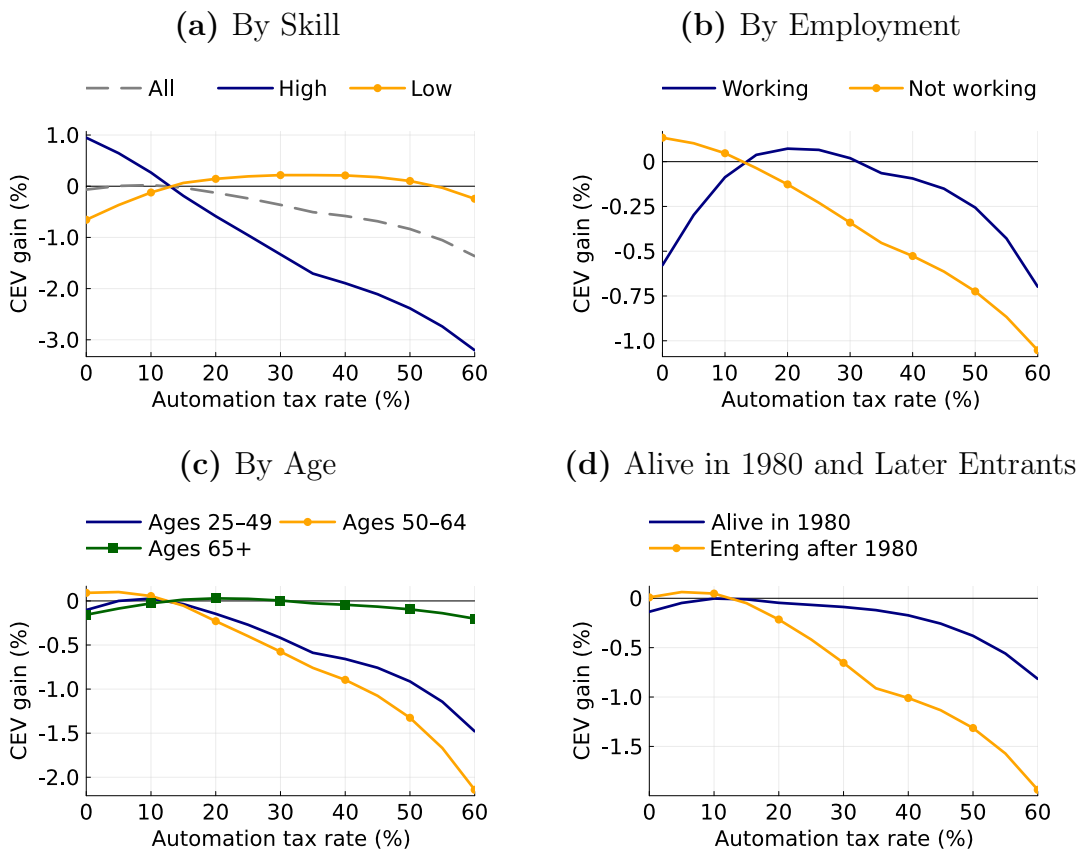
Notes: This table shows the calibrated utility costs of occupational choice. The upper rows show the mobility cost $c^s_{oo'}$ of moving from the occupation on the left to the occupation above. The bottom row shows the occupation-specific cost $\kappa_{s,o'}$ of working in each occupation, normalized to zero for routine occupations. Costs are in units of utility. Mobility costs match occupation transition probabilities of CPS household heads aged 30–59 in 1983–1985, and occupation-specific costs match occupation shares among employed individuals at age 25 in 1983–1985. Low-skilled individuals have at most a high school education; high-skilled individuals have at least some college education.

Table A.2: Occupational Choice Components of the 2018 Entry Cohort

	Automation only		Full polarization	
	Low	High	Low	High
Occupation costs	−6.32	−1.73	−8.41	−1.04
Switching costs	−5.57	1.37	−4.48	3.47
Preference shocks	12.83	0.19	9.68	−5.15
Occupational choice (net)	0.94	−0.16	−3.22	−2.73

Notes: This table shows welfare changes for 2018 entrants by skill from occupation-specific costs, switching costs and preference-shock utility, and their sum, the net value of occupational choice, with automation only and with full polarization, each relative to constant technology. Entries are $100\Delta U_k(y,s)/[(1-\gamma)U^b_c(y,s)]$, where $\Delta U_k(y,s) = U^a_k(y,s) - U^b_k(y,s)$ is the utility change in component k for entry cohort y and skill s, a denotes the alternative economy, b the constant-technology benchmark, and $U^b_c(y,s)$ its consumption utility; positive entries raise welfare, including when they represent reduced costs. Preference-shock utility is the expected preference shock of the chosen alternative, accumulated over the life cycle with survival and discounting.

Figure A.1: Welfare Effects by Automation Tax Rate: Transfers to Workers



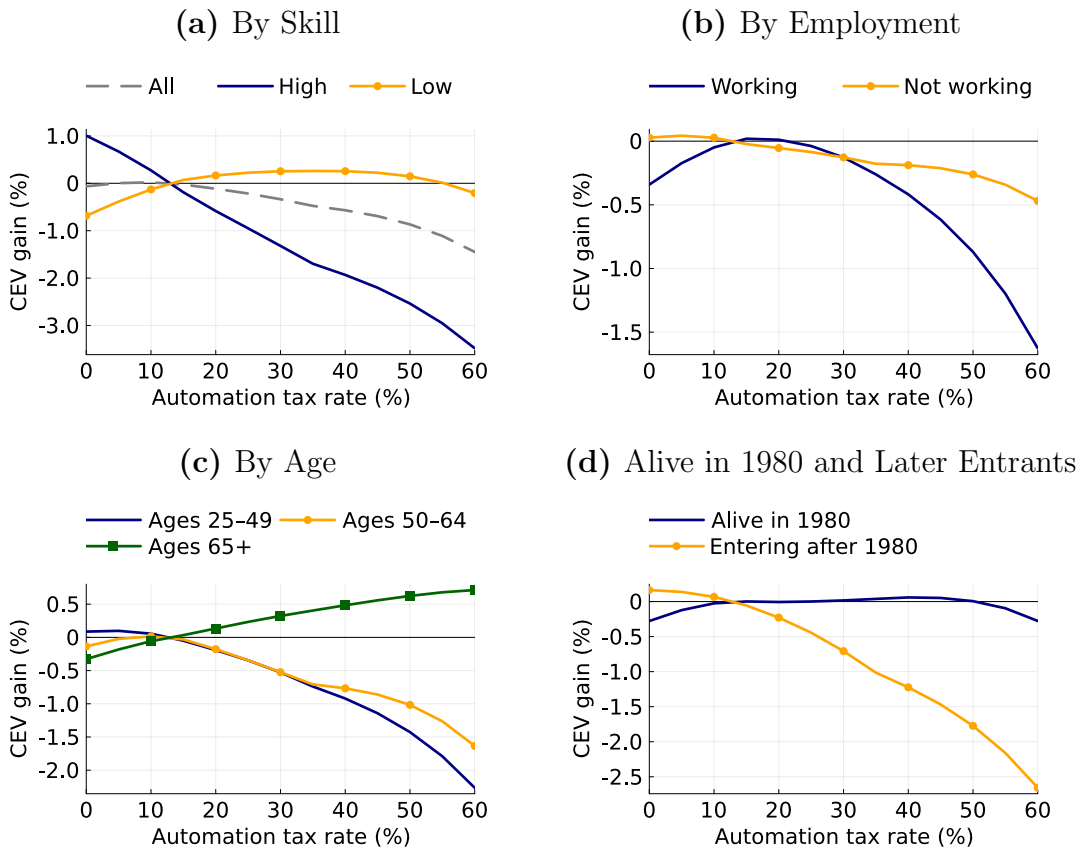
Notes: This figure shows CEV gains relative to the baseline, in which the automation tax is 13%, for constant automation tax rates from 0% to 60% in 5-point steps, imposed from 1981, when the change in revenue is paid each year as an equal transfer only to individuals who work in that year, in abstract, routine, or manual occupations. Individuals who do not work receive nothing, and households know the rule when they choose whether to work. The transfer balances the government budget each year; the labor income tax stays at 25.59% and government expenditures follow the baseline path. Panel (a) shows social welfare (All) and the same welfare for high-skilled (High) and low-skilled (Low) individuals. Panels (b) and (c) group each year's population by its state in that year, so an individual moves between groups over the life cycle: by employment in panel (b) and by age in panel (c). Each group's utility in each year from 1981 is discounted to 1980 as in social welfare, and the gain compares utility per member and year, since employment responds to the tax. Panel (d) splits social welfare into individuals alive in 1980 and later entrants. Gains are in percentage terms; the status quo reproduces baseline welfare up to numerical precision.

Table A.3: Economies by Automation Tax Rate: Transfers to Workers

		Automation tax rate (%); status quo: 13%				
		0	10	20	30	40
CEV gain (%)						
All		−0.07	0.02	−0.13	−0.36	−0.58
High		0.95	0.27	−0.59	−1.33	−1.89
Low		−0.66	−0.12	0.14	0.22	0.21
Working		−0.58	−0.09	0.07	0.02	−0.09
Not working		0.13	0.05	−0.13	−0.34	−0.53
Ages 25–49		−0.10	0.03	−0.15	−0.42	−0.66
Ages 50–64		0.09	0.06	−0.23	−0.58	−0.90
Ages 65+		−0.16	−0.03	0.03	0.00	−0.04
Alive in 1980		−0.14	0.00	−0.05	−0.09	−0.17
Entering after 1980		0.01	0.05	−0.22	−0.65	−1.01
Output (%)	2018	7.1	1.7	−4.2	−10.6	−12.8
	2300	6.2	1.5	−3.6	−9.2	−12.9
High wage (%)	2018	1.2	0.3	−0.7	−1.8	−2.9
	2300	0.9	0.2	−0.6	−1.5	−2.4
Low wage (%)	2018	−0.9	−0.2	0.5	1.6	1.2
	2300	−1.0	−0.3	0.7	1.8	2.1
Routine wage (%)	2018	−1.6	−0.4	1.0	2.8	2.7
	2300	−1.8	−0.4	1.1	2.9	3.8
Routine share (pp)	2018	−3.8	−0.9	2.2	5.1	5.9
	2300	−3.6	−0.9	2.1	5.3	7.4
Non-employment share (pp)	2018	1.0	0.2	−0.5	−0.7	−0.8
	2300	0.9	0.2	−0.5	−1.2	−1.7
Automation share (pp)	2018	6.7	1.6	−3.7	−8.6	−10.0
	2300	6.7	1.6	−3.8	−9.2	−12.5

Notes: This table shows the economy under constant automation tax rates imposed from 1981 relative to the status quo, in which the tax is 13%; CEV gains compare welfare under each policy with welfare under the status quo; annual outcomes are differences from the status quo in the same year. The change in revenue is paid each year as an equal transfer only to individuals who work in that year, in abstract, routine, or manual occupations; households know the rule when they choose whether to work. The labor income tax stays at 25.59% and government expenditures follow the baseline path. CEV gains are for social welfare (All), which weights individuals alive in 1980 and future entrants discounted to 1980, and the same welfare for high-skilled (High) and low-skilled (Low) individuals; each year's population grouped by employment and by age in that year; and individuals alive in 1980 and later entrants. Output and wages are percentage differences: output, the high- and low-skill wages, which are wage indices that weight log occupational wages by the 1980 earnings shares of the high-skilled and the low-skilled, and the routine wage. Shares are differences in percentage points (pp): the share of routine occupations among employed individuals aged 30–59, the share of individuals aged 30–59 who are not employed, and the share of automated tasks. 2300 is the final stationary economy.

Figure A.2: Welfare Effects by Automation Tax Rate: Labor Market Rents



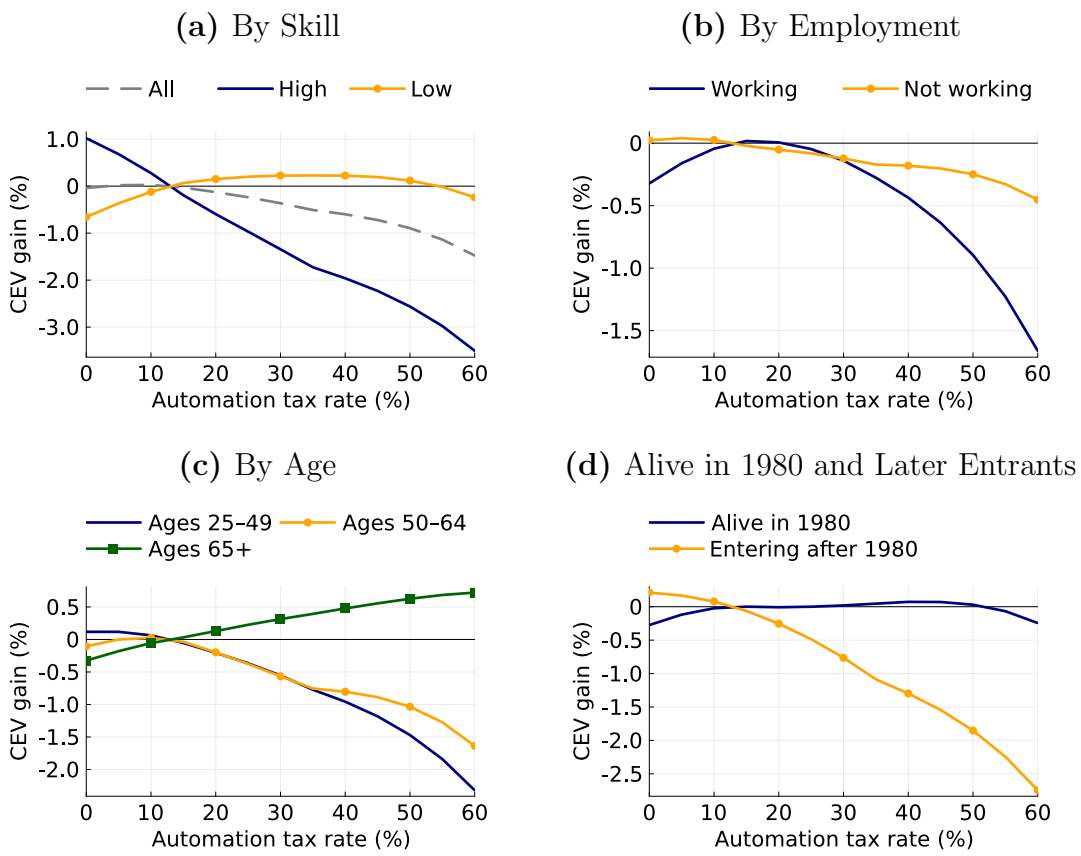
Notes: This figure shows CEV gains relative to the baseline, in which the automation tax is 13%, for constant automation tax rates from 0% to 60% in 5-point steps, imposed from 1981, when each year of work adds to welfare the rent $\varrho(1-\tau_{I,t})y$ on earnings y, with $\varrho = 7.5\%$. Allocations are those of the main specification, in which the change in revenue is paid as a uniform lump sum to everyone alive. Panel (a) shows social welfare (All) and the same welfare for high-skilled (High) and low-skilled (Low) individuals. Panels (b) and (c) group each year's population by its state in that year, so an individual moves between groups over the life cycle: by employment in panel (b) and by age in panel (c). Each group's utility in each year from 1981 is discounted to 1980 as in social welfare, and the gain compares utility per member and year, since employment responds to the tax. Panel (d) splits social welfare into individuals alive in 1980 and later entrants. Gains are in percentage terms; the status quo reproduces baseline welfare up to numerical precision.

Table A.4: Welfare Gains by Automation Tax Rate: Labor Market Rents

	Automation tax rate (%); status quo: 13%				
	0	10	20	30	40
CEV gain (%), $\varrho = 7.5\%$					
All	−0.06	0.02	−0.11	−0.34	−0.57
High	1.00	0.27	−0.58	−1.32	−1.93
Low	−0.68	−0.13	0.17	0.25	0.26
Working	−0.34	−0.05	0.01	−0.13	−0.42
Not working	0.03	0.03	−0.05	−0.13	−0.19
Ages 25–49	0.09	0.06	−0.19	−0.53	−0.92
Ages 50–64	−0.14	0.01	−0.18	−0.52	−0.77
Ages 65+	−0.33	−0.06	0.13	0.32	0.48
Alive in 1980	−0.28	−0.03	−0.01	0.02	0.06
Entering after 1980	0.16	0.07	−0.23	−0.71	−1.22
CEV gain (%), $\varrho = 15\%$					
All	−0.08	0.01	−0.10	−0.31	−0.54
High	0.98	0.27	−0.57	−1.29	−1.89
Low	−0.69	−0.13	0.18	0.27	0.27
Working	−0.34	−0.05	0.01	−0.13	−0.42
Not working	0.03	0.03	−0.05	−0.13	−0.19
Ages 25–49	0.07	0.05	−0.18	−0.51	−0.91
Ages 50–64	−0.17	0.00	−0.15	−0.47	−0.71
Ages 65+	−0.33	−0.06	0.14	0.33	0.49
Alive in 1980	−0.27	−0.03	−0.01	0.01	0.04
Entering after 1980	0.12	0.06	−0.21	−0.65	−1.15

Notes: This table shows CEV gains for constant automation tax rates imposed from 1981 relative to the status quo, in which the tax is 13%, when each year of work adds to welfare the rent $\varrho(1 - \tau_{I,t})y$ on earnings y, with $\varrho = 7.5\%$ and $\varrho = 15\%$. Allocations are those of the main specification, in which the change in revenue is paid as a uniform lump sum to everyone alive. CEV gains are for social welfare (All), which weights individuals alive in 1980 and future entrants discounted to 1980, and the same welfare for high-skilled (High) and low-skilled (Low) individuals; each year's population grouped by employment and by age in that year; and individuals alive in 1980 and later entrants.

Figure A.3: Welfare Effects by Automation Tax Rate: Manual Labor in the CES Aggregate



Notes: This figure shows CEV gains relative to the baseline, in which the automation tax is 13%, for constant automation tax rates from 0% to 60% in 5-point steps, imposed from 1981, when manual labor enters the composite X_t with abstract labor and routine tasks at elasticity σ. The automation technology, the baseline paths and the equilibrium at each rate are solved under this specification. The change in revenue is paid as a uniform lump sum to everyone alive; the labor income tax stays at 25.59% and government expenditures follow the baseline path. Panel (a) shows social welfare (All) and the same welfare for high-skilled (High) and low-skilled (Low) individuals. Panels (b) and (c) group each year's population by its state in that year, so an individual moves between groups over the life cycle: by employment in panel (b) and by age in panel (c). Each group's utility in each year from 1981 is discounted to 1980 as in social welfare, and the gain compares utility per member and year, since employment responds to the tax. Panel (d) splits social welfare into individuals alive in 1980 and later entrants. Gains are in percentage terms; the status quo reproduces baseline welfare up to numerical precision.

Table A.5: Economies by Automation Tax Rate: Manual Labor in the CES Aggregate

		Automation tax rate (%); status quo: 13%				
		0	10	20	30	40
CEV gain (%)						
All		−0.04	0.03	−0.13	−0.36	−0.60
High		1.02	0.28	−0.60	−1.34	−1.96
Low		−0.65	−0.12	0.15	0.23	0.22
Working		−0.32	−0.04	0.01	−0.14	−0.44
Not working		0.02	0.03	−0.05	−0.12	−0.18
Ages 25–49		0.12	0.06	−0.20	−0.55	−0.96
Ages 50–64		−0.10	0.02	−0.20	−0.57	−0.80
Ages 65+		−0.32	−0.06	0.13	0.31	0.48
Alive in 1980		−0.27	−0.02	−0.01	0.02	0.07
Entering after 1980		0.21	0.08	−0.25	−0.76	−1.30
Output (%)	2018	7.2	1.7	−4.2	−10.5	−12.6
	2300	6.2	1.5	−3.6	−9.2	−12.9
High wage (%)	2018	1.2	0.3	−0.7	−1.8	−2.9
	2300	0.9	0.2	−0.6	−1.5	−2.4
Low wage (%)	2018	−0.9	−0.2	0.5	1.5	1.2
	2300	−1.1	−0.3	0.7	1.8	2.1
Routine wage (%)	2018	−1.6	−0.4	1.0	2.7	2.7
	2300	−1.8	−0.4	1.1	2.9	3.8
Routine share (pp)	2018	−3.7	−0.9	2.2	5.1	5.9
	2300	−3.5	−0.8	2.1	5.3	7.4
Non-employment share (pp)	2018	0.8	0.2	−0.4	−0.8	−0.9
	2300	0.7	0.2	−0.5	−1.2	−1.7
Automation share (pp)	2018	6.5	1.5	−3.7	−8.6	−10.0
	2300	6.5	1.6	−3.7	−9.2	−12.5

Notes: This table shows the economy under constant automation tax rates imposed from 1981 relative to the status quo, in which the tax is 13%, when manual labor enters the composite X_t with abstract labor and routine tasks at elasticity σ. The change in revenue is paid as a uniform lump sum to everyone alive; the labor income tax stays at 25.59% and government expenditures follow the baseline path. CEV gains are for social welfare (All), which weights individuals alive in 1980 and future entrants discounted to 1980, and the same welfare for high-skilled (High) and low-skilled (Low) individuals; each year's population grouped by employment and by age in that year; and individuals alive in 1980 and later entrants. Output and wages are percentage differences: output, the high- and low-skill wages, which are wage indices that weight log occupational wages by the 1980 earnings shares of the high-skilled and the low-skilled, and the routine wage. Shares are differences in percentage points (pp): the share of routine occupations among employed individuals aged 30–59, the share of individuals aged 30–59 who are not employed, and the share of automated tasks. 2300 is the final stationary economy.